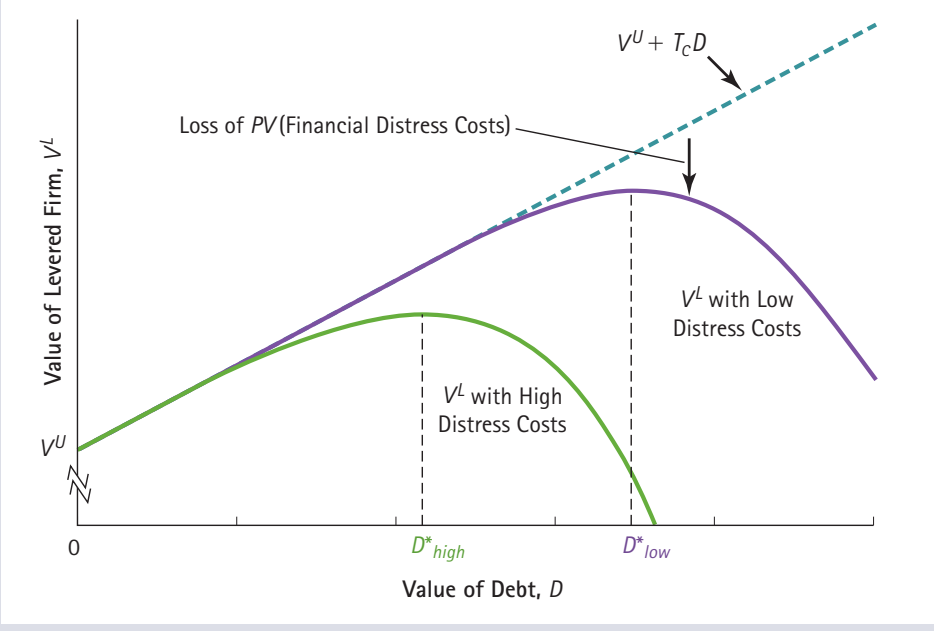
Berk

Fundamentals of Corporate Finance, 2nd Edition {Jonathan Ber

- Agency costs of leverage: asset substitution and debt overhang: Ch. 16, §16.6 (p. 482–487)

FIGURE 16.8
Optimal Leverage with Taxes and Financial Distress Costs

As the level of debt increases, the tax benefits of debt increase until the interest expense exceeds the firm's EBIT. However, the probability of default, and hence the present value of financial distress costs, also increases. The optimal level of debt, D^*, occurs when these effects balance out and the value of the levered firm is maximized. D^* will be lower for firms with higher costs of financial distress.



optimal debt choice for a firm with high costs of financial distress is indicated by D^*_{high}. Not surprisingly, with higher costs of financial distress, it is optimal for the firm to choose lower leverage.

The tradeoff theory helps to resolve two important facts about leverage:

1. The presence of financial distress costs can explain why firms choose debt levels that are too low to fully exploit the interest tax shield.
2. Differences in the magnitude of financial distress costs and the volatility of cash flows can explain the differences in the use of leverage across industries.

Concept Check

9. According to the tradeoff theory, how should a financial manager determine the right capital structure for a firm?
10. Why would managers in one industry choose different capital structures than those in another industry?

16.6 Additional Consequences of Leverage: Agency Costs and Information

Taxes and financial distress costs are not the only capital market imperfections that arise in practice. In this section, we continue to relax the assumption of perfect capital markets to see other ways in which capital structure can affect a firm's value. We begin by discussing how leverage alters managers' incentives and changes their investment decisions. We then address the complexities of stakeholders in the firm having different information.

Agency Costs

agency costs The costs that arise when there are conflicts of interest between stakeholders.

Agency costs are costs that arise when there are conflicts of interest between stakeholders. In Chapter 1, we briefly mentioned the agency problem that arises when managers put their own interests ahead of shareholders' interest. Here, we discuss how debt can mitigate this problem. However, it may also distort equity holders' preferences for the types of projects pursued by the firm.

Managerial Entrenchment. Although managers often do own shares of the firm, in most large corporations they own only a very small fraction of the outstanding shares. Shareholders, through the board of directors, have the power to fire managers. In practice, they rarely do so unless the firm's performance is exceptionally poor.

management entrenchment A situation arising as a result of the separation of ownership and control, in which managers may make decisions that benefit themselves at investors' expense.

This separation of ownership and control creates the possibility of **management entrenchment**; facing little threat of being fired and replaced, managers are free to run the firm in their own best interests. As a result, managers may make decisions that benefit themselves at investors' expense. Managers may reduce their effort, spend excessively on perks such as corporate jets, or undertake wasteful projects that increase the size of the firm (and their paychecks) at the expense of shareholder value, often called "empire building." If these decisions have a negative NPV for the firm, they are a form of agency cost.

These agency costs are most likely to arise when equity ownership is highly diluted (so that no individual shareholder has an incentive to monitor management closely) and when the firm has a great deal of cash available for managers to spend on wasteful projects. Debt can therefore help in two ways. First, by borrowing rather than raising funds by issuing shares, ownership of the firm may remain more concentrated, improving the monitoring of management. Second, by forcing the firm to pay out cash to meet interest and principal payments, debt reduces the funds available at management's discretion. For managers to engage in wasteful investment, they must have the cash to invest. Only when cash is tight will managers be motivated to run the firm as efficiently as possible. Thus, leverage can provide incentives for managers to run the firm efficiently and effectively. These benefits provide an additional incentive to use debt rather than equity financing.

Airlines Use Financial Distress to Their Advantage

The need to generate cash flows sufficient to make interest payments may also tie managers' hands and commit them to pursue sound business strategies with greater vigor than they would without the threat of financial distress. For example, when American Airlines was in labor negotiations with its unions in April 2003, the firm was able to win wage concessions by explaining that higher costs would push it into bankruptcy. (A similar situation enabled Delta Airlines to persuade its pilots to accept a 33% wage cut in November 2004.) Without the threat of financial distress, American's managers might not have reached agreement with the union as quickly or achieved the same wage concessions.

A firm with greater leverage may also become a fiercer competitor and act more aggressively in protecting its markets because it cannot risk the possibility of bankruptcy. This commitment to aggressive behavior can scare off potential rivals. (However, this argument can also work in reverse, as a firm weakened by too much leverage might become so financially fragile that it crumbles in the face of competition, allowing other firms to erode its markets.)

Equity-Debt Holder Conflicts. When a firm has leverage, a conflict of interest exists if investment decisions have different consequences for the value of equity and the value of debt. Such a conflict is most likely to occur when the risk of financial distress is high. In some circumstances, managers may take actions that benefit shareholders but harm the firm's creditors and lower the total value of the firm.

To illustrate, put yourself in the place of the shareholders of a company in distress that is likely to default on its debt. You could continue as normal, in which case you will very likely lose the value of your shares and control of the firm to the bondholders.

Alternatively, you could:

1. Roll the dice and take on a risky project that could save the firm, even though its expected outcome is so poor that you normally would not take it on.
2. Conserve funds rather than invest in new, promising projects.
3. Cash out by distributing as much of the firm's capital as possible to the shareholders before the bondholders take over.

Why would you roll the dice? Even if the project fails, you are no worse off because you were headed to default anyway. If it succeeds, then you avoid default and retain ownership of the firm. This incentive leads to **excessive risk-taking**, a situation that occurs when a company is near distress and shareholders have an incentive to invest in risky negative-NPV projects that will destroy value for debt holders and the firm overall. Anticipating this behavior, security holders will pay less for the firm initially. This cost is likely to be highest for firms that can easily increase the risk of their investments.

excessive risk-taking A situation that occurs when a company is near distress and shareholders have an incentive to invest in risky negative-NPV projects that will destroy value for debt holders and the firm overall.

On the other hand, when default is very likely, a firm may pass up a good project before default because some or most of the benefit will go to bondholders (by reducing the risk or extent of default). For example, if a firm can invest \$100 in a project that will increase the value of its debt by \$30 and the value of its equity by \$90, then the project has a positive NPV of \$30 + \$90 − \$100 = \$20. Equity holders will not want to fund it as their gain (\$90) is less than the project's cost of \$100. In this case, there is an **under-investment problem**: Shareholders choose not to invest in a positive-NPV project because the firm is in financial distress and the value of undertaking the investment opportunity will accrue to bondholders rather than themselves. This failure to invest is costly for debt holders and for the overall value of the firm, because it is giving up the NPV of the missed opportunities. The cost is highest for firms that are likely to have profitable future growth opportunities requiring large investments.

under-investment problem A situation in which shareholders choose not to invest in a positive-NPV project because the firm is in financial distress and the value of undertaking the investment opportunity will accrue to bondholders rather than to shareholders.

The ultimate form of under-investment is cashing out. Knowing that they are likely to lose the firm to the bondholders, shareholders have an incentive to withdraw as much capital from the firm as possible before it enters bankruptcy and transfers to the debt holders. An extreme example of this would be to sell all the assets of the firm and distribute the proceeds to shareholders as a special dividend. The firm would then enter bankruptcy as a worthless shell. As we discussed in Chapter 15, bondholders can anticipate this problem and often require restrictions on the size of dividends and source of funds for those dividends.

Our discussion suggests that we can create a more complete tradeoff model of capital structure by incorporating the possible agency costs and benefits of debt in addition to tax benefits and financial distress costs. This tradeoff model representing optimal leverage is illustrated in Figure 16.9. When debt is too low, adding leverage increases the value of the firm by providing tax benefits and motivating managers to run the firm more efficiently (and avoid wasteful investment). But if debt is too high, the firm will incur financial distress costs and suffer from excessive risk taking and under-investment. The optimal level of debt D^* is the point that maximizes firm value by balancing these positive and negative consequences of leverage.

Financial Distress and Rolling the Dice, *Literally*

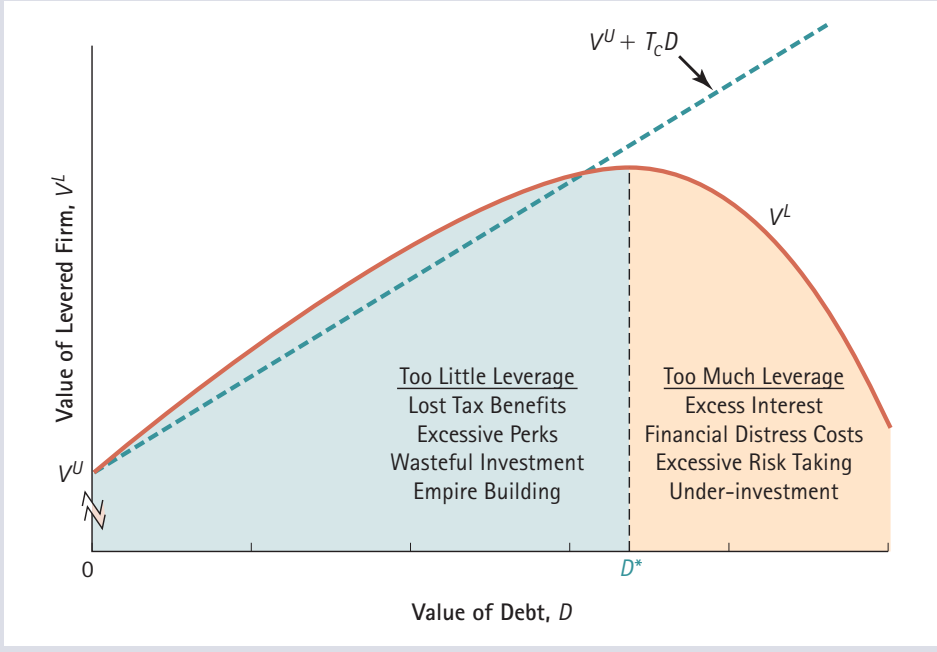
Shortly after launching FedEx, founder Frederick Smith was running out of cash and on the verge of shuttering the company's operations. The problem was massive fixed costs (airplanes, sorting facilities, delivery personnel, etc.) and low initial volume. "By 1973, Smith was so desperate for cash that he flew to Las Vegas to play the blackjack tables. He wired the \$27,000 he won back to FedEx."* The company used the winnings to pay its fuel bills, and the rest is history.

*"Frederick W. Smith: No Overnight Success," *Business Week,* September 20, 2004.

FIGURE 16.9

Optimal Leverage with Taxes, Financial Distress, and Agency Costs

As the level of debt increases, the value of the firm increases from the interest tax shield ($T_C D$) as well as improvements in managerial incentives. If leverage is too high, however, the present value of financial distress costs, as well as the agency costs from debt holder–equity holder conflicts, dominates and reduces firm value. The optimal level of debt, D^*, balances these benefits and costs of leverage.



Debt and Information

asymmetric information A situation in which parties have different information. It can arise when, for example, managers have superior information to that of outside investors regarding the firm's future cash flows.

signaling theory of debt The use of leverage as a way to signal good information to investors.

market timing When managers sell new shares because they believe the stock is overvalued, and rely on debt and retained earnings (and possibly repurchasing shares) if they believe the stock is undervalued.

Our final market imperfection is related to the role of information. Throughout this chapter, we have assumed that managers, stockholders, and creditors have the same information. We have also assumed that securities are fairly priced: The firm's shares and debt are priced according to their true underlying value. These assumptions may not always be accurate in practice. Due to **asymmetric information**, managers' information about the firm and its future cash flows is likely to be superior to that of outside investors. This asymmetric information may motivate managers to alter a firm's capital structure.

Leverage as a Credible Signal. To convince the stock market that it has great projects, a firm may commit to large future debt payments. If the projects are great, then the firm will have no trouble making the debt payments. But if the firm is making false claims and then does not grow, it will have trouble paying its creditors and will experience financial distress. This distress will be costly for the firm and also for its managers, who will likely lose their jobs. Thus, the managers can use leverage as a way to convince investors that they do have information that the firm will grow, even if they cannot provide verifiable details about the sources of growth. Investors will interpret the additional leverage as a credible signal of the managers' confidence. The use of leverage as a way to signal good information to investors is known as the **signaling theory of debt**.

Market Timing. When managers have better information than outside investors regarding the value of the firm, they may attempt to engage in **market timing** by selling new shares when they believe the stock is overvalued, and relying on debt and retained

earnings (and possibly repurchasing shares) if they believe the stock is undervalued. Managers who successfully time the market in this way benefit long-term shareholders by trading the firm's shares whenever they are mispriced. The firm's capital structure would change as the managers issued new equity (decreasing the leverage) or new debt (increasing the leverage) while attempting to time the market. As a result, the firm's capital structure may deviate above or below the optimal level described by the tradeoff theory, depending on management's view of the share price relative to its true value.

Adverse Selection and the Pecking Order Hypothesis. Suppose managers do try to issue equity when it is overpriced relative to its true value. Knowing this, how would investors react? Recall from Chapter 14 the adverse selection or lemons effect in seasoned equity offerings. Fearful that they are being sold a "lemon," investors will discount the price they are willing to pay for the stock.

The adverse selection problem has implications for capital structure. Managers do not want to sell equity if they have to discount it to find buyers. Therefore, they may seek alternative forms of financing. Debt issues may also suffer from adverse selection. Because the value of low-risk debt is not very sensitive to managers' private information about the firm (but is instead determined mainly by interest rates), the discount needed to attract buyers will be smaller for debt than for equity. Of course, a firm can avoid underpricing altogether by financing investment using its cash (retained earnings) when possible. The **pecking order hypothesis** therefore states that to avoid this "lemons cost":

pecking order hypothesis The idea that managers will have a preference to fund investment using retained earnings, followed by debt, and will only choose to issue equity as a last resort.

> *Managers will have a preference to fund investment using retained earnings, followed by debt, and will only choose to issue equity as a last resort.*

This hypothesis has implications for firms' capital structure. When firms are profitable and generate sufficient cash to fund their investments, they will not issue debt or equity, but just rely on retained earnings. Thus, highly profitable firms will have little debt in their capital structure. Only firms that need to raise external capital will have significant debt financing. According to the pecking order hypothesis, firms should almost never issue equity. In reality, it is likely that multiple forces will shape the firm's capital structure, and firms will issue equity when the agency costs or financial distress costs of debt are too great.

EXAMPLE 16.5 The Pecking Order of Financing Alternatives

Problem

Axon Industries needs to raise $9.5 million for a new investment project. If the firm issues one-year debt, it may have to pay an interest rate of 8%, although Axon's managers believe that 6% would be a fair rate given the level of risk. However, if the firm issues equity, the managers believe the equity may be underpriced by 5%. What is the cost to current shareholders of financing the project out of retained earnings, debt, and equity?

Solution

Plan

We can evaluate the financing alternatives by comparing what the firm would have to pay to get the financing versus what its managers believe it should pay if the market had the same information they do.

Execute

If the firm spends $9.5 million out of retained earnings, rather than paying that money out to shareholders as a dividend, the cost of financing the project is $9.5 million.

Using one-year debt costs the firm $\$9.5 \times (1.08) = \$10.26$ million in one year, which has a present value based on management's view of the firm's risk of $\$10.26 \div (1.06) = \$9.68$ million.

If equity is underpriced by 5%, then to raise $9.5 million the firm will need to issue shares that are actually worth $10 million. (For example, if the firm's shares are each worth $50, but it sells them for

0.95 × $50 = $47.50 per share, it will need to sell $9.5 million ÷ $47.50/share = 200,000 shares. These shares have a true value of 200,000 shares × $50/share = $10 million.) Thus, the cost of financing the project with equity will be $10 million.

Evaluate

Comparing the three options, retained earnings are the cheapest source of funds, followed by debt, and finally by equity. The ranking reflects the effect of differences in information between managers and investors that result in a lemons problem when they issue new securities, and particularly when they issue new equity.

Concept Check

11. How can too much debt lead to excessive risk-taking?
12. What is the "pecking order" hypothesis?

16.7 Capital Structure: Putting It All Together

In this chapter, we have examined a number of factors that might influence a firm's choice of capital structure. What is the bottom line for a financial manager devising the optimal capital structure for a firm? The optimal capital structure depends on market imperfections such as taxes, financial distress costs, agency costs, and asymmetric information, as follows:

1. *Make use of the interest tax shield if your firm has consistent taxable income.* The interest tax shield allows firms to repay investors and avoid the corporate tax.
2. *Balance the tax benefits of debt against the costs of financial distress when determining how much of the firm's income to shield from taxes with leverage.* Whereas the risk of default is not itself a problem, financial distress may lead to other consequences that reduce the value of the firm.
3. *Consider short-term debt for external financing when agency costs are significant.* Too much debt can motivate managers and equity holders to take excessive risks or under-invest in a firm. When free cash flows are high, too little leverage may encourage wasteful spending.
4. *Increase leverage to signal managers' confidence in the firm's ability to meet its debt obligations.* Investors understand that bankruptcy is costly for managers.
5. *Be mindful that investors are aware that you have an incentive to issue securities that you know are overpriced.* Thus, when an issue is announced, investors will lower their valuation of that security. This effect is most pronounced for equity issues because the value of equity is most sensitive to the manager's private information.
6. *Rely first on retained earnings, then debt, and finally equity.* This pecking order of financing alternatives will be most important when managers are likely to have a great deal of private information regarding the value of the firm.
7. *Do not change the firm's capital structure unless it departs significantly from the optimal level.* Actively changing a firm's capital structure (for example, by selling or repurchasing shares or bonds) entails transactions costs. Most changes to a firm's debt-equity ratio are likely to occur passively, as the market value of the firm's equity fluctuates with changes in the firm's stock price.

20. Suppose Kohwe borrows the $50 million instead. The firm will pay interest only on this loan each year, and it will maintain an outstanding balance of $50 million on the loan. Suppose that Kohwe's corporate tax rate is 40%, and expected free cash flows are still $10 million each year. What is Kohwe's share price today if the investment is financed with debt?

21. Now suppose that with leverage, Kohwe's expected free cash flows will decline to $9 million per year due to reduced sales and other financial distress costs. Assume that the appropriate discount rate for Kohwe's future free cash flows is still 8%. What is Kohwe's share price today given the financial distress costs of leverage?

Optimal Capital Structure: The Tradeoff Theory

22. Hawar International is a shipping firm with a current share price of $5.50 and 10 million shares outstanding. Suppose that Hawar announces plans to lower its corporate taxes by borrowing $20 million and repurchasing shares, that Hawar pays a corporate tax rate of 30%, and that shareholders expect the change in debt to be permanent.
    a. If the only imperfection is corporate taxes, what will the share price be after this announcement?
    b. Suppose the only imperfections are corporate taxes and financial distress costs. If the share price rises to $5.75 after this announcement, what is the PV of financial distress costs Hawar will incur as the result of this new debt?

23. Marpor Industries has no debt and expects to generate free cash flows of $16 million each year. Marpor believes that if it permanently increases its level of debt to $40 million, the risk of financial distress may cause it to lose some customers and receive less favorable terms from its suppliers. As a result, Marpor's expected free cash flows with debt will be only $15 million per year. Suppose Marpor's tax rate is 35%, the risk-free rate is 5%, the expected return of the market is 15%, and the beta of Marpor's free cash flows is 1.10 (with or without leverage).
    a. Estimate Marpor's value without leverage.
    b. Estimate Marpor's value with the new leverage.

Additional Consequences of Leverage: Agency Costs and Information

24. Dynron Corporation's primary business is natural gas transportation using its vast gas pipeline network. Dynron's assets currently have a market value of $150 million. The firm is exploring the possibility of raising $50 million by selling part of its pipeline network and investing the $50 million in a fiber-optic network to generate revenues by selling high-speed network bandwidth. Whereas this new investment is expected to increase profits, it will also substantially increase Dynron's risk. If Dynron is levered, would this investment be more or less attractive to equity holders than if Dynron had no debt?

25. Consider a firm whose only asset is a plot of vacant land, and whose only liability is debt of $15 million due in one year. If left vacant, the land will be worth $10 million in one year. Alternatively, the firm can develop the land at an upfront cost of $20 million. The developed land will be worth $35 million in one year. Suppose the risk-free interest rate is 10%, assume all cash flows are risk-free, and assume there are no taxes.
    a. If the firm chooses not to develop the land, what is the value of the firm's equity today? What is the value of the debt today?
    b. What is the NPV of developing the land?

c. Suppose the firm raises $20 million from equity holders to develop the land. If the firm develops the land, what is the value of the firm's equity today? What is the value of the firm's debt today?
d. Given your answer to part (c), would equity holders be willing to provide the $20 million needed to develop the land?

*26. Zymase is a biotechnology start-up firm. Researchers at Zymase must choose one of three different research strategies. The payoffs (after taxes) and their likelihood for each strategy are shown below. The risk of each project is diversifiable.

Strategy	Probability	Payoff ($ million)
A	100%	75
B	50%	140
	50%	0
C	10%	300
	90%	40

a. Which project has the highest expected payoff?
b. Suppose Zymase has debt of $40 million due at the time of the project's payoff. Which strategy has the highest expected payoff for equity holders?
c. Suppose Zymase has debt of $110 million due at the time of the strategy's payoff. Which strategy has the highest expected payoff for equity holders?
d. If management chooses the strategy that maximizes the payoff to equity holders, what is the expected agency cost to the firm from having $40 million in debt due? What is the expected agency cost to the firm from having $110 million in debt due?

*27. You own a firm, and you want to raise $30 million to fund an expansion. Currently, you own 100% of the firm's equity, and the firm has no debt. To raise the $30 million solely through equity, you will need to sell two-thirds of the firm. However, you would prefer to maintain at least a 50% equity stake in the firm to retain control.
a. If you borrow $20 million, what fraction of the equity will you need to sell to raise the remaining $10 million? (Assume perfect capital markets.)
b. What is the smallest amount you can borrow to raise the $30 million without giving up control? (Assume perfect capital markets.)

*28. Empire Industries forecasts net income this coming year as shown below (in thousands of dollars):

EBIT	$1,000
Interest expense	0
Income before tax	1,000
Taxes	−350
Net income	$650

Approximately $200,000 of Empire's earnings will be needed to make new, positive-NPV investments. Unfortunately, Empire's managers are expected to waste 10% of its net income on needless perks, pet projects, and other expenditures that do not contribute to the firm. All remaining income will be distributed to shareholders.
a. What are the two benefits of debt financing for Empire?
b. By how much would each $1 of interest expense reduce Empire's distributions to shareholders?
c. What is the increase in the *total* funds Empire will pay to investors for each $1 of interest expense?

Grinblatt

Financial Markets And Corporate Strategy{Mark Grinblatt, She

- Short-term vs long-term debt as a cure for debt overhang: Ch. 16, §16.4 (p. 584–586)

risky firms, which are less likely to be subject to the types of financial distress costs discussed in this chapter. For these firms, the benefits of bank debt are insufficient to justify the added costs associated with bank loans—costs that are passed on to the borrower in the form of higher interest payments.

There are a number of costs associated with bank debt that firms may be able to avoid by going directly to the bond or commercial paper market. The first is what we would call a pure intermediation cost that arises because banks require buildings and labor and are unable to loan out all their funds because of reserve requirements. A second cost has to do with potential incentive problems within banks that arise because loan officers may extend additional credit to marginal borrowers either because they don't want to reveal that their initial loan to the firm was bad or because they don't want to face the unpleasant task of forcing a client into bankruptcy. For the reasons discussed earlier, the costs associated with a bank's suboptimal actions may need to be passed on to the firm borrowing from it. Finally, there may exist what is known as a hold up problem that makes bank debt less attractive for some borrowers. A **hold up problem** occurs if a firm relies too much on any single bank, and the bank takes advantage of this reliance and charges the firm above-market rates on its subsequent loans. In such a situation, the firm may find it difficult to access capital from other, potentially cheaper, sources, since the firm's inability to raise the capital from its bank could be viewed as a signal that the bank believes the firm's prospects are not favorable.[22]

The Use of Short-Term versus Long-Term Debt

Most of the debt holder–equity holder conflicts discussed in this chapter are more severe when firms use long-term rather than short-term debt financing. To understand why this is true, recall that these conflicts arise because equity holders have an incentive to implement investment strategies that are advantageous to them by lowering the value of the firm's outstanding debt. Hence, since the value of short-term debt is much less sensitive to changes in a firm's investment strategy than the value of long-term debt, conflicts are lower for firms financed with short-term debt.

Mitigating the Debt Overhang Problem. Myers (1977) noted that it is possible to eliminate the debt overhang problem if the firm's existing debt matures before the time when it must raise additional debt to fund a new project. To understand this, consider again the example of Unitron, which had a risk-free project yielding 12 percent when the risk-free rate was 10 percent. The firm passed up the project because its borrowing costs were high enough to exceed the project's return. If the firm's debt was all short-term, there would have been no debt overhang problem. In this case, the interest payment on all of the firm's debt would have been renegotiated simultaneously with the selection of the new project. Thus, the addition of a risk-free project, which would lower the overall risk of the firm, also would lower the firm's borrowing costs, making it attractive for the firm to accept the project.

Mitigating the Asset Substitution Problem. The use of short-term debt also makes it more difficult for equity holders to gain at the expense of debt holders by selecting riskier projects. Consider the case of a firm that must decide whether to design

[22]See Rajan (1992) for a discussion of the hold up problem.

its production facilities so that the process is more or less risky. The equity holders of a highly leveraged firm may prefer the risky project because the upside potential is higher and they share any downside risk with the debt holders. However, if the firm's debt is primarily short term, the incentive to increase asset risk diminishes. With short-term financing, the lending rate is renegotiated after completion of the production process, which largely eliminates a firm's ability to gain at the expense of its lenders.

Disadvantages of Short-Term Debt. Of course, there also is a downside to funding long-term projects with short-term debt. With short-term financing, unexpected increases in interest rates could potentially bankrupt a highly leveraged firm. This has led some authors to advocate short-term borrowing coupled with hedging interest rate risk in the futures and swap markets.[23]

Security Design: The Use of Convertibles

A convertible bond provides its holder with the option to exchange bonds for a prespecified number of the issuing firm's shares (see Chapter 2). Some analysts view a convertible bond as a combination of a call option on the firm's stock and a straight bond. While this description is not altogether correct, it does provide useful intuition. Recall that options, and thus the option element in the convertible bond, become more valuable as the volatility of the firm's stock increases. This increase in the value of the option component can offset the decrease in the value of the convertible's straight bond component that occurs when the firm's volatility increases.

Designing a Convertible Bond to Make Its Value Insensitive to Volatility Changes. Depending on the relative importance of the option and straight bond components, the convertible bond can either increase or decrease in value with an increase in the firm's overall level of risk. Some researchers have suggested that it may be possible to design a convertible bond so that its value is insensitive to changes in the volatility of the firm.[24] The equity holders of a firm financed with such debt would therefore have no incentive to select high risk projects since they would gain nothing from the debt holders and would not profit from a project's adoption unless the project has a positive *NPV*.

Empirical Evidence on Convertible Issuance. Empirical evidence by Mikkelson (1981) tends to support this rationale for convertible bond issuance. Mikkelson found that highly leveraged, high-growth firms were the most likely to issue convertible bonds. These firms are likely to have the highest probability of bankruptcy, so the effect of risk on the value of their straight bonds and stock is probably the greatest. In addition, these firms are likely to have the greatest flexibility when it comes to future investments and thus the greatest ability to increase risk. Mikkelson also found that the maturities of the convertible bonds issued by firms were generally longer than the maturities of their straight bonds. This probably reflects the inability of growth firms, because of incentive problems, to obtain long-term financing with straight debt.

[23] These issues are discussed in greater detail in Chapter 21.
[24] See Brennan and Schwartz (1986) and Green (1984).

The Use of Project Financing

Project financing is capital to finance an investment project for which both the project's assets and the liabilities attached to its financing can effectively be separated from the rest of the firm. For example, most major U.S. oil companies structure at least some of their foreign operations as separate operating units with their own financing. Similarly, power generating plants are often structured as independently financed units of electrical utilities.[25] In most cases, the parent company provides equity for the project (in many cases along with a joint venture partner), which appears on the firm's balance sheet. However, the parent is not responsible for the project's debt (that is, the project is financed with what is called **non-recourse debt**), which implies that the debt need not appear on the parent firm's balance sheet. Moreover, the project's debt has a senior claim on the cash flows generated by the project.

The use of project financing can mitigate the debt holder–equity holder conflicts in a number of ways. Recall the case of Lily Pharmaceutical, considered earlier, where a positive *NPV* project was passed up because the wealth transferred from equity holders to debt holders exceeded the project's *NPV*. In this case, if project financing were used, there would be substantially less of a wealth transfer from equity holders to debt holders because the project's debt would have the senior claim on the project's cash flows. Moreover, since project financing is tied to a specific project, there is generally less scope for the kind of asset substitution problems discussed in Section 16.2.

Of course, project financing is not a panacea that solves all debt holder–equity holder conflicts. In many cases, it is difficult to define a firm's project in a way that allows it to be financed as a separate entity. Suppose, for example, that Ford can upgrade its domestic manufacturing facilities at a cost of $3 billion. Since the manufacturing facilities are already owned by Ford, financing their renovation would be difficult to structure as a separate project.

In addition, although project financing, when it can be used, generally mitigates the underinvestment problem, it can often exacerbate the asset substitution problem. To understand why project financing can exacerbate the asset substitution problem consider the behavior of an entrepreneur with ten diverse investment projects that can be financed either as ten independent projects or together as one firm. If each of the ten projects is financed with non-recourse debt, the incentive to increase the risk of any specific project may be quite high since the entrepreneur will capture all the upside benefits associated with the more favorable outcomes while the debt holders bear the increased downside costs associated with the unfavorable outcomes. However, if the projects are all financed under the umbrella of one corporation, with conventional debt financing, part of the gain associated with the favorable outcomes in any given project benefits the debt holders since it might offset an unfavorable outcome from another project. Put differently, because of diversification, the value of debt is not particularly sensitive to the fortunes or misfortunes of any one project since most of the project-specific risk has been diversified away.

[25]The Hollywood film *Erin Brockovich* describes how project financing provided a measure of protection from lawsuits for Pacific Gas and Electric. For more in-depth information about project financing we recommend a book by Finnerty (1996) and an article by Kensinger and Martin (1988).

rate is 4 percent greater than the risk-free rate. After reviewing your recommendation, the company CEO has asked you to explain how this "positive net present value project" can make him money when he is forced to borrow at 12 percent to fund a project yielding 10 percent. You wonder how you bungled an assignment as simple as evaluating a risk-free project. What have you done wrong?

16.10. In the event of bankruptcy, the control of a firm passes from the equity holders to the debt holders. Describe differences in the preferences of the equity holders and debt holders and how decisions following bankruptcy proceedings are likely to change.

16.11. Why are debt holder–equity holder incentive problems less severe for firms that borrow short term rather than long term?

16.12. Consider the case of Ajax Manufacturing which just completed an R&D project on widgets that required a $70 million bond obligation. The R&D effort resulted in an investment opportunity that will cost $75 million and generate cash flows of $85 million in the event of a recession (*prob.* = 20%) and $150 million if economic conditions are favorable (*prob.* = 80%). What is the *NPV* of the project assuming no taxes, no direct bankruptcy costs, risk neutrality, and a risk-free interest rate of zero? Can the firm fund the project if the original debt is a senior obligation that doesn't allow the firm to issue additional debt?

16.13. Assume now that if Ajax Manufacturing (see exercise 16.12) uses a more capital-intensive manufacturing process, it can produce a greater number of widgets at a lower variable cost. Given the greater fixed costs, the cash flows are only $5 million in an unfavorable economy with the capital-intensive process but are $170 million in a favorable economy. Hence, equity holders would receive $100 million in the good state of the economy ($170 million − $70 million) and zero in a recession because $5 million is less than the $70 million debt obligation. Can the firm issue equity to fund the project?

16.14. With debtor in possession (DIP) financing, bankrupt firms are able to obtain additional amounts of debt that is senior to the firm's existing debt. Explain how the firm's existing debt holders can benefit from this.

16.15. You have been hired as a bond analyst for Bull Sterns. A highly leveraged firm, Emax Industries, has switched to a more flexible management process that enables it to change its investment strategy more quickly. How do you expect this change in the management process to affect bond values?

16.16. Atways Industries is involved in two similar mining projects. The Wyoming project was financed through the firm's internal cash flows and appears as an asset on its balance sheet. The Montana project was set up as a wholly owned subsidiary of Atways. The subsidiary was financed 20 percent with equity provided by Atways and 80 percent with non-recourse debt.

How do the different ways that these projects were originally financed and structured affect future investment and operating decisions?

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Ross

ROSS, WESTERFIELD, JORDAN_ LAMB - Fundamentos de Administraç

- O problema de agência e o controle da empresa: Cap. 1, §1.4 (p. 12–17)

biliários (CVM) brasileira exige que as divulgações realizadas pelas empresas listadas no Brasil tragam as mesmas informações em todos os mercados (se fosse diferente, acionistas em alguns mercados teriam informações mais completas ou diferentes das fornecidas para acionistas em outros mercados), o padrão de divulgação no mercado brasileiro, das empresas brasileiras listadas nos Estados Unidos, também acaba seguindo o padrão SOX.

Para essa convergência de práticas de divulgação, contribui também o fato de que as grandes firmas de auditoria que atuam no Brasil têm atuação global e se submetem à fiscalização do PCAOB – *Public Company Accounting Oversight Board*, instituído pela SOX para fiscalizar as firmas de auditoria (ver em http://pcaobus.org/Pages/default.aspx). Isso faz que suas práticas de auditoria estejam alinhadas às exigências da SOX, o que acaba refletindo nas práticas de auditoria no Brasil.

Questões conceituais

1.3a Qual é o objetivo da administração financeira?

1.3b Quais são algumas das desvantagens do objetivo de maximizar os lucros?

1.3c Você consegue fornecer uma definição de finanças corporativas?

1.4 O problema de agência e o controle da empresa

Já vimos que o administrador financeiro busca o melhor interesse dos acionistas, adotando medidas que aumentam o valor das ações. Entretanto, também vimos que, em grandes corporações, a propriedade pode se diluir em um número enorme de acionistas. Essa dispersão de propriedade, sem dúvida, significa que quem realmente controla a empresa é a sua administração. Nesse caso, a administração agirá necessariamente para buscar o melhor interesse dos acionistas? Em outras palavras, a administração não poderia buscar seus próprios objetivos em detrimento dos acionistas? Nas próximas páginas, faremos uma consideração rápida sobre alguns dos argumentos relacionados a essa questão.

Relacionamentos de agência

O relacionamento entre acionistas e administradores é chamado de *relacionamento de agência*. Tal relacionamento existe sempre que alguém (o principal) contrata outra pessoa (o agente) para representar seus interesses. Por exemplo, você pode contratar alguém (um agente, representante seu) para vender um carro seu enquanto você está na escola. Nesses relacionamentos, existe a possibilidade de conflitos de interesses entre principal e agente. Tal conflito é um **problema de agência**.

problema de agência
a possibilidade de conflitos de interesses entre os acionistas e a administração de uma empresa.

Suponhamos que você contrate alguém para vender seu carro e que concorda em pagar a essa pessoa uma comissão fixa quando o carro for vendido. O incentivo do representante, nesse caso, é fazer a venda, não necessariamente conseguir o melhor preço. Se você oferecer uma comissão de, por exemplo, 10% sobre o preço da venda em vez de uma comissão fixa, então provavelmente não haverá esse problema. Esse exemplo mostra que a maneira como um agente é pago afeta os problemas de agência.[†]

Objetivos da administração

Para ver como os interesses de administradores e acionistas podem ser diferentes, imagine que a empresa esteja considerando um novo investimento, que deve ter impacto favorável sobre o valor das ações mas que também é um empreendimento relativamente arriscado. Os proprietários da empresa vão querer fazer o investimento (porque o valor da ação subirá), mas os administradores podem não querer, porque existe a possibilidade de que as coisas não deem certo e

[†] N. de R. T.: Já se tornaram usuais nos textos do Brasil os termos "principal" e "agente" no contexto da "teoria da agência".

eles percam seus empregos. Se a administração não aceitar o investimento, os acionistas podem perder uma oportunidade valiosa. Este é um exemplo de um custo de agência.

De modo mais geral, o termo *custo de agência* se refere aos custos do conflito de interesses entre acionistas e administradores. No contexto brasileiro, a administração da maior parte das empresas de capital aberto é exercida por acionistas controladores definidos; então o conflito de agência no Brasil, e os custos decorrentes, são derivados principalmente da possível divergência entre acionistas controladores e acionistas não controladores. Por exemplo, controladores podem se apropriar de "rendas de controle" que podem resultar de escolhas decisórias que beneficiam partes relacionadas aos controladores, ou que derivam da apropriação de benefícios privados, como empréstimos da empresa aos controladores, aeronaves de uso dos controladores com os custos pagos pela empresa aberta, etc.

Os custos de agência podem ser diretos ou indiretos. Um custo de agência indireto é uma oportunidade perdida, como a que acabamos de descrever.

Os custos diretos de agência são de dois tipos. O primeiro é um gasto da empresa que beneficia a administração, mas que custa para os acionistas. A compra de um jatinho luxuoso e desnecessário pode ser vista como um gasto desse tipo. O segundo tipo é uma despesa que surge da necessidade de monitorar as ações dos administradores. O pagamento de auditores externos para avaliar a exatidão das informações das demonstrações financeiras seria um exemplo disso.

Às vezes, argumenta-se que, se deixados por conta própria, os administradores tenderiam a maximizar os recursos sobre os quais têm controle ou, de maneira geral, o poder ou a riqueza da empresa. Esse objetivo levaria a uma ênfase excessiva no tamanho ou no crescimento da empresa. Por exemplo, são comuns os casos em que uma administração é acusada de realizar gastos muito superiores ao valor de mercado na compra de outra empresa, apenas para aumentar o tamanho da empresa que controlam ou para demonstrar poder empresarial. Obviamente, se o gasto exagerado ocorrer realmente, tal compra não beneficiará os acionistas da empresa compradora.

Nossa discussão indica que, no caso de empresas dominadas pelos gestores, a administração pode tender a enfatizar excessivamente a sobrevivência organizacional para garantir os seus empregos. Além disso, a administração pode não gostar de interferência externa, de modo que a independência e a autossuficiência corporativa podem ser objetivos importantes para os administradores desse tipo de empresa.

Os administradores buscam os interesses dos acionistas?

Nas empresas controladas por gestores que não são acionistas controladores, o fato de os administradores realmente buscarem o melhor interesse dos acionistas vai depender de dois fatores. Em primeiro lugar, os objetivos dos administradores estão mesmo alinhados aos objetivos dos acionistas? Essa pergunta se relaciona, pelo menos em parte, ao modo como os administradores são remunerados. Em segundo lugar, a administração pode ser substituída se não buscar os objetivos dos acionistas? Essa pergunta se relaciona ao controle da empresa. Como discutiremos, existem vários motivos para pensar que, mesmo nas grandes empresas, os administradores têm um incentivo significativo para buscar o melhor interesse dos acionistas.

Como afirmamos anteriormente, no caso brasileiro, de um modo geral os acionistas controladores exercem a direção e gestão dos negócios, se não na qualidade de executivos, pelo controle do conselho de administração e, de perto, pelo controle do grupo de executivos. Nessa situação, o conflito entre administração e acionista tende a ser mínimo; o conflito potencial maior é de outra natureza, é o que pode ocorrer entre acionistas controladores e acionistas não controladores, como mencionamos antes.

Remuneração dos administradores É comum que a administração tenha incentivos econômicos significativos para aumentar o valor das ações por dois motivos. Primeiramente, a remuneração dos executivos, particularmente os do primeiro escalão, está ligada não só ao desempenho financeiro, mas também quase sempre ao valor da ação da companhia. Por exemplo, os administradores frequentemente recebem opções para comprar ações a preços de barganha. Quanto mais valer a ação, mais valiosa é essa opção. Na verdade, as opções são muito utilizadas para

motivar os funcionários de todos os tipos, e não apenas os administradores de primeiro escalão. Por exemplo, em 2007, o Google anunciou a emissão de novas opções de ações para 16 mil funcionários, dando à sua força de trabalho uma participação significativa no preço de mercado de suas ações e alinhando os interesses dos funcionários aos dos acionistas. Muitas outras corporações, grandes e pequenas, têm adotado políticas semelhantes.

O segundo incentivo que os administradores têm se relaciona às perspectivas de emprego. Os administradores com melhor desempenho tendem a ser promovidos. De forma geral, os administradores bem-sucedidos na busca dos objetivos dos acionistas serão mais procurados pelo mercado de trabalho e, assim, poderão exigir maiores salários.

Na verdade, os administradores bem-sucedidos nesse quesito podem receber enormes recompensas. Por exemplo, um dos executivos mais bem pagos em 2007 foi Lawrence Ellison, o CEO da Oracle; de acordo com a revista Forbes, ele recebeu cerca de $193 milhões. A título de comparação, Ellison recebeu menos que J.K. Rowling ($300 milhões) e Oprah Winfrey ($275 milhões), mas muito mais que Tiger Woods ($115 milhões). No período de 2003 a 2007, o CEO da Apple, Steve Jobs, foi o executivo mais bem pago, ganhando em tomo de $661 milhões. Informações sobre a remuneração dos executivos, e muitas outras informações sobre quase todas as empresas abertas, podem ser encontradas facilmente na Internet. A seção Exercícios na Internet mostra como fazer isso. No caso brasileiro, esta informação estará disponível no formulário de referência de forma agregada entre valor mínimo e máximo separando a remuneração do conselho e da diretoria executiva válida somente para exercício.



Questões de ética nos negócios são abordadas em www.business-ethics.com.

Controle da empresa Em última instância, o controle da empresa está nas mãos dos acionistas. Eles elegem o conselho de administração que, por sua vez, contrata e demite os administradores. O fato de os acionistas controlarem a corporação ficou bastante claro pela experiência de Steve Jobs na Apple. Embora ele tenha sido o fundador da empresa e responsável pelos produtos de maior sucesso, houve um momento em que os acionistas, por meio de seus conselheiros eleitos, resolveram que a Apple estaria melhor sem ele e, assim, ele foi mandado embora. Posteriormente, ele foi recontratado e ajudou a empresa a recuperar-se com novos produtos, tais como o iPod e o iPhone.

Nos Estados Unidos, um mecanismo importante que os acionistas descontentes podem usar para substituir a administração é chamado de *disputa por procurações de voto*. A procuração de voto é uma autorização para votar com ações de outro. Uma disputa por procurações de voto se desenvolve quando um grupo faz ofertas agressivas por procurações para substituir o conselho existente e, portanto, substituir os administradores existentes. Por exemplo, no início de 2002, a proposta de incorporação entre a Hewlett-Packard (HP) e a Compaq deflagrou uma das disputas por procurações de voto mais divulgadas, amargas e caras da história, atingindo bem mais de $100 milhões. Um grupo de acionistas, incluindo Walter B. Hewlett (membro do conselho e herdeiro de um cofundador da HP), se opôs à incorporação e iniciou uma disputa por procurações de voto para assumir o controle da HP. Outro grupo, liderado pela CEO da HP, Carly Fiorina, apoiava a incorporação. Em uma votação muito apertada, Fiorina venceu, a incorporação prosseguiu e Hewlett renunciou ao cargo no conselho de administração.

No Brasil, disputas por procurações não são o caminho usual para afastar a direção de uma empresa pelo simples fato de que raramente as ações com direito a voto têm suficiente dispersão no mercado para permitir isso. Mesmo no caso das empresas listadas no Novo Mercado (para as quais a totalidade das ações emitidas tem poder de voto) observa-se a existência de blocos de controle majoritários, tornando inútil a busca de votos que se revelarão insuficientes para mudar decisões nas assembleias de acionistas.

Outra maneira de substituir a administração é por meio de aquisição da empresa. Empresas mal administradas são mais atraentes como aquisições do que as bem administradas, porque o potencial de lucro é maior. Assim, evitar uma aquisição por outra empresa é algo que dá à administração um incentivo para agir no interesse dos acionistas. Por exemplo, em 2008, o portal Web Yahoo! estava lutando contra uma oferta pública indesejada de aquisição de $42 bilhões feita pela gigante dos *software* Microsoft. A administração do Yahoo! alegou que a oferta subestimava significativamente o valor da empresa, embora fosse 62% maior que o então preço

corrente das suas ações. Em uma tentativa de frustrar a tentativa de aquisição, o Yahoo! fechou um contrato de publicidade com o rival *site* de buscas Google e começou a discutir com outras empresas, tais como a AOL, sobre uma possível fusão.

No Brasil, uma forma de buscar o alinhamento de interesses entre acionistas e a gestão das empresas listadas foi a instituição dos segmentos especiais de listagem de ações da BM&FBOVESPA, segregados por níveis de governança corporativa.

O Novo Mercado e os níveis diferenciados de governança corporativa criados pela BM&FBOVESPA[†]





Para saber mais sobre o Novo Mercado e os níveis diferenciados de governança corporativa, visite o site www.bmfbovespa.com.br/empresas/pages/empresas_segmentos-de-listagem.asp

Os segmentos especiais de listagem do mercado de ações (Novo Mercado, Nível 2, Nível 1 e BOVESPA Mais) foram criados pela BM&FBOVESPA há mais de 10 anos, no momento em que a Bolsa percebeu que, para desenvolver o mercado de capitais brasileiro, atraindo novos investidores e novas empresas, era preciso ter segmentos de listagem com regras rígidas de governança corporativa. No Capítulo 15 você vai encontrar mais informações sobre os níveis de governança relacionados ao momento da oferta pública de ações pela empresa.

Essas regras vão além das obrigações que as companhias têm perante a Lei das Sociedades por Ações (Lei das S/As.) e melhoram a avaliação das companhias que decidem aderir, voluntariamente, a um desses níveis de listagem.

Além disso, regras mais rígidas reduzem o risco dos investidores que decidem ser sócios destas empresas, graças aos direitos e garantias asseguradas aos acionistas e a divulgação de informações mais completas, que reduzem as assimetrias de informações entre acionistas controladores, gestores da companhia e os participantes do mercado.

As empresas listadas no Novo Mercado da BM&FBOVESPA devem ter elevados padrões de governança corporativa; elas também só podem emitir ações com direito de voto, as chamadas ações ordinárias nominativas (ON).

Por se tratar de uma mudança relevante na estrutura de capital das empresas, é mais comum que a decisão de integrar o segmento especial de listagem do Novo Mercado seja tomada quando da abertura de capital.

Principais regras do Novo Mercado relacionadas à estrutura de governança e aos direitos dos acionistas

- o capital deve ser composto exclusivamente por ações ordinárias com direito a voto;
- no caso de venda do controle todos os acionistas têm direito a vender suas ações pelo mesmo preço (*tag along* de 100%);
- em caso de deslistagem ou cancelamento do contrato do Novo Mercado com a BM&FBOVESPA, a empresa deverá fazer oferta pública para recomprar as ações de todos os acionistas no mínimo pelo valor econômico;
- o conselho de administração deve ser composto por no mínimo cinco membros, sendo 20% dos conselheiros independentes e todos os conselheiros devem ter mandato máximo de dois anos;
- a companhia também se compromete a manter no mínimo 25% das ações em circulação no mercado (*free float*);
- divulgação de dados financeiros deve ser mais completa que a exigida em lei, incluindo relatórios trimestrais com demonstração de fluxo de caixa e relatórios consolidados revisados por um auditor independente;
- a empresa deverá disponibilizar relatórios financeiros anuais em um padrão internacionalmente aceito;
- a empresa deve divulgar mensalmente as negociações com valores mobiliários da companhia por diretores, executivos e acionistas controladores.

[†] N. de E.: Material cedido pelo Instituto Educacional BM&FBOVESPA (www.bmfbovespa.com.br/educacional).

O **Nível 2** exige que as companhias aceitem e cumpram todas as obrigações previstas no regulamento do Novo Mercado, com algumas exceções. As empresas listadas no Nível 2 têm o direito de manter ações preferenciais nominativas (PN). No caso de venda de controle da empresa, é assegurado aos detentores de ações preferenciais o direito de *tag along*, no mínimo, de 80% do preço pago pelas ações ordinárias do acionista controlador (no Novo Mercado essa exigência é de 100%).

As ações preferenciais ainda dão direito de voto aos acionistas em situações críticas, como a aprovação de fusões e incorporações da empresa e contratos entre o acionista controlador e a companhia, sempre que essas decisões estiverem sujeitas à aprovação na assembleia de acionistas.

O **Nível 1** exige que as empresas adotem práticas que favoreçam a transparência e o acesso às informações pelos investidores. Para isso, devem divulgar informações adicionais às exigidas em lei, como relatórios financeiros mais completos, informações sobre negociação feita por diretores, executivos e acionistas controladores e sobre operações com partes relacionadas. Companhias listadas nesse segmento devem também manter um *free float* mínimo de 25%.

O segmento especial de listagem **BOVESPA Mais** foi idealizado pela BM&FBOVESPA para tornar o mercado de ações brasileiro acessível a um número maior de empresas, especialmente àquelas que desejam entrar no mercado aos poucos, como empresas de pequeno e médio porte, que consideram o mercado de capitais uma importante fonte de recursos e que buscam adotar estratégias diferentes de ingresso: captação de volumes menores, aumento de exposição junto ao mercado para criar valor e realização de distribuições de valores mobiliários mais concentradas, por exemplo.

As empresas listadas no BOVESPA Mais tendem a atrair investidores que percebem nelas um potencial de desenvolvimento mais acentuado, quando comparadas com empresas listadas no mercado principal.

As regras de listagem do BOVESPA Mais são semelhantes às do Novo Mercado e, da mesma forma, as empresas nele listadas assumem compromissos de elevados padrões de governança corporativa e transparência com o mercado.

A Tabela 1.2 resume as principais diferenças entre os segmentos especiais de listagem do mercado de ações na BM&FBOVESPA.

TABELA 1.2 Principais diferenças entre os segmentos especiais de listagem do mercado de ações na BM&FBOVESPA

	BOVESPA Mais	Novo mercado	Nível 2	Nível 1	Tradicional
Percentual mínimo de ações em circulação (*free float*)	25% de *free float* até o sétimo ano de listagem ou condições mínimas de liquidez	No mínimo 25% de *free float*	No mínimo 25% de *free float*	No mínimo 25% de *free float*	Não há regra
Características das ações emitidas	Somente ações ON podem ser negociadas e emitidas, mas é permitida a existência de PN	Permite a existência somente de ações ON	Permite a existência de ações ON e PN (com direitos adicionais)	Permite a existência de ações ON e PN	Permite a existência de ações ON e PN
Conselho de administração	Mínimo de três membros (conforme legislação)	Mínimo de cinco membros, dos quais pelo menos 20% devem ser independentes	Mínimo de cinco membros, dos quais pelo menos 20% devem ser independentes	Mínimo de três membros (conforme legislação)	Mínimo de três membros (conforme legislação)
Concessão de *Tag Along*	100% para ações ON	100% para ações ON	100% para ações ON 80% para ações PN	80% para ações ON (conforme legislação)	80% para ações ON (conforme legislação)
Adoção da câmara de arbitragem do mercado	Obrigatório	Obrigatório	Obrigatório	Facultativo	Facultativo

Conclusão A teoria e as evidências disponíveis são coerentes com a visão de que os acionistas controlam a empresa e de que a maximização da riqueza do acionista é o objetivo mais impor-

tante da empresa. Mesmo assim, sem dúvida há momentos em que os objetivos dos administradores tomam a dianteira em detrimento dos acionistas, ao menos temporariamente.

Públicos de relacionamento (*stakeholders*)

Nossa discussão até aqui faz parecer que a administração e os acionistas são as únicas partes que têm interesse nas decisões da empresa. Sem dúvida, essa é uma visão muito simplificada dos fatos. Funcionários, clientes, fornecedores e até mesmo o governo têm interesse financeiro na empresa.

Juntos, esses diversos grupos são chamados de **públicos de relacionamento**. Em geral, uma parte interessada é alguém que não é acionista ou credor e que tem um interesse potencial sobre os fluxos de caixa da empresa. Tais grupos também tentarão exercer controle sobre a empresa, talvez em detrimento dos proprietários.

público de relacionamento (*stakeholder*)
alguém que não é acionista ou credor e que tem um interesse potencial sobre os fluxos de caixa da empresa.

Questões conceituais

1.4a O que é um relacionamento de agência?

1.4b O que são problemas de agência e como eles surgem? O que são custos de agência?

1.4c Que incentivos os administradores de corporações grandes recebem para maximizar o valor das ações?

1.5 Mercados financeiros e a empresa

Já vimos que as principais vantagens da sociedade por ações como forma de organização de empresas são que a propriedade pode ser transferida de forma mais rápida e fácil do que em outras formas de organização e que é possível angariar dinheiro com maior facilidade. As duas vantagens são ampliadas significativamente pela existência de mercados financeiros, os quais desempenham um papel muito importante nas finanças corporativas.

Fluxos de caixa na empresa

A interação entre a sociedade por ações e os mercados financeiros é ilustrada na Figura 1.2. As setas da figura indicam a passagem do dinheiro dos mercados financeiros para a empresa e dela para os mercados financeiros.

Suponhamos que uma empresa inicia vendendo ações e tomando dinheiro emprestado para levantar fundos. O dinheiro flui dos mercados financeiros para a empresa (A). A empresa investe o dinheiro em ativos circulantes e permanentes (B). Esses ativos geram dinheiro (C), parte do qual vai para o pagamento dos impostos da pessoa jurídica (D). Após o pagamento dos impostos, parte desse fluxo de caixa é reinvestida na empresa (E). O restante volta para os mercados financeiros como dinheiro pago aos credores e acionistas (F).

Um mercado financeiro, assim como qualquer mercado, é apenas uma forma de reunir compradores e vendedores. Nele, títulos de dívida e ações são comprados e vendidos. Entretanto, os mercados financeiros diferem nos detalhes. As diferenças mais comuns dizem respeito aos tipos de títulos negociados, a como a negociação é conduzida e a quem são os compradores e os vendedores. Algumas dessas diferenças são discutidas a seguir.

Mercados primários *versus* mercados secundários

Os mercados financeiros funcionam como mercados primários e secundários para os títulos de dívida e ações. O termo *mercado primário* refere-se à venda original de títulos pelos governos e pelas empresas. Os *mercados secundários* são aqueles nos quais esses títulos são comprados e vendidos após a venda original. As ações, obviamente, só são emitidas pelas empresas. Os títulos da dívida são emitidos pelos governos e pelas empresas. Na discussão a seguir, nossa atenção recai apenas sobre os títulos de empresas.

4. Em uma grande empresa, existe a possibilidade de conflitos entre acionistas e administradores e entre acionistas controladores e não controladores. Chamamos esses conflitos de *problemas de agência* e discutimos como eles podem ser controlados e reduzidos.
5. As vantagens da sociedade por ações são ampliadas pela existência dos mercados financeiros. Eles funcionam como mercados primário e secundário para os títulos corporativos e podem ser organizados como mercados de balcão e de bolsa.

Dos tópicos que discutimos até agora, o mais importante é o que trata do objetivo da administração financeira: maximizar o valor das ações. Ao longo deste livro, analisaremos muitas decisões financeiras diferentes, mas sempre faremos a mesma pergunta: como tal decisão afeta o valor das ações?

REVISÃO DE CONCEITOS E QUESTÕES INSTIGANTES

1. **O processo de decisão da administração financeira [OA1]** Quais são os três tipos de decisões da administração financeira? Para cada tipo de decisão, dê exemplo de uma transação de negócios pertinente.
2. **Empresas individuais e sociedades [OA3]** O que é uma Eireli? Quais são as principais desvantagens da empresa individual e de uma sociedade? Quais benefícios existem nesses tipos de organização de negócios quando comparados à forma de sociedade por ações? Descreva as diferenças entre as sociedades: em nome coletivo; em comandita simples; limitada; em comandita por ações.
3. **Sociedades por ações [OA3]** Cite pelo menos duas vantagens da sociedade por ações.
4. **Sarbanes-Oxley [OA4]** Em resposta a lei Sarbanes-Oxley, muitas pequenas empresas optaram por "sumir dos holofotes" e tirar suas ações da bolsa. Por que uma empresa deveria escolher esse caminho? Quais são os custos de "sumir dos holofotes"?
5. **Organização financeira da empresa [OA4]** Em uma grande empresa, quais são os dois grupos distintos que se subordinam ao diretor financeiro? Qual grupo é o foco das finanças corporativas?
6. **Objetivo da administração financeira [OA2]** Qual objetivo deve sempre motivar as ações do administrador financeiro da empresa?
7. **Problemas de agência [OA4]** Quem tem a propriedade de uma organização? Descreva o processo pelo qual os proprietários controlam a administração da empresa. Qual é o principal motivo para a existência de um relacionamento de agência na forma sociedade por ações de organização? Nesse contexto, que tipos de problemas podem surgir?
8. **Mercados primários *versus* mercados secundários [OA3]** Provavelmente você já notou a cobertura na imprensa especializada de uma oferta pública inicial (Initial Public Offering – IPO) dos títulos mobiliários de uma empresa. Uma IPO é uma transação do mercado primário ou secundário?
9. **Mercados de bolsa *versus* mercados de balcão [OA3]** O que significa dizer que a Bolsa de Valores de Nova York é um mercado de pregão? Em que os mercados de pregão são diferentes dos mercados de balcão? Que tipo de mercado é a Nasdaq? Que tipo de mercado é a BM&FBOVESPA?
10. **Objetivos de empresas sem fins lucrativos [OA2]** Suponhamos que você seja o administrador financeiro de uma empresa sem fins lucrativos (um hospital beneficiente, por exemplo). Que objetivos você acha que seriam apropriados?
11. **Objetivo da empresa [OA2]** Avalie a seguinte declaração: os administradores não devem se concentrar no valor atual da ação, porque isso levará a uma ênfase excessiva nos lucros de curto prazo em detrimento dos lucros de longo prazo.
12. **Ética e objetivos da empresa [OA2]** Nosso objetivo de maximizar o valor das ações pode entrar em conflito com outros objetivos. Nesse caso, como evitar o comportamento antiético ou ilegal? Você acha que assuntos como segurança do cliente e do funcionário,

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Tirole

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- Fixed-investment model: IC constraint, pledgeable income, credit rationing: Ch. 3, §3.2 (p. 115–125)
- Variable-investment model: equity multiplier: Ch. 3, §3.4 (p. 127–130)
- Diversification and cross-pledging: Ch. 4, §4.2 (p. 158–164)

of the assets in the case of failure. We show that it is optimal for investors to have priority in the case of default and for the entrepreneur to be the residual claimant (Chapter 5 will investigate another feature of debt, namely, the borrower's promise to pay fixed amounts to investors as a going concern, i.e., before liquidation; and Chapter 10 will connect debtholders' control rights with their cash-flow rights).

By focusing on a simple model of credit rationing, we do not do full justice to the corporate finance literature, which has developed a wide array of models with a similar flavor. For the sake of completeness, the supplementary section studies three broad classes of models that, through more sophisticated modeling, have aimed at deriving an interpretation of the "leftover claim" of outside investors as a standard debt claim.

3.2 The Role of Net Worth: A Simple Model of Credit Rationing

3.2.1 The Fixed-Investment Model

Variants of the following entrepreneurial model[2] will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. (We will later explore the possibility that these assets be illiquid. For example, they might be equipment or premises that are needed for the implementation of the project.) The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. (We will later observe that we can ignore the possibility that the entrepreneur consumes some of the cash and borrows more than $I - A$.)

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit");[3] or, equivalently, the entrepreneur chooses between a project with a high probability of success and another project which ceteris paribus she prefers (is easier to implement, is more fun, has greater spinoffs in the future for the entrepreneur, benefits a friend, delivers perks, is more "glamorous," etc.) but has a lower probability of success.[4]

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur.[5] In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral.[6] For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0.[7] The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. That is, we have in mind that several prospective lenders compete for issuing a loan to the borrower, and that, if the most attractive loan offer made a positive profit, the borrower could turn to an alternative lender and offer

2. This specific model is taken from Holmström and Tirole (1997). But its main idea can be found in various forms in many anterior papers.

3. See Exercise 3.20 for the continuous-effort version of the model.

4. Note that, for simplicity, we treat the entrepreneur as a unitary actor. There is an interesting question as to how moral hazard and incentives propagate down within the corporate hierarchy. Pagano and Volpin (2005) assume that benefits accrue to all company insiders, and not only to managers; in their model, managers need workers' cooperation to produce and therefore share benefits with employees.

5. For example, in the biotechnology alliance financing discussed in Section 2.4.2, the private benefit might be the entrepreneur's benefit from working on other projects (with other partners or on her own). The shift in attention then reduces the probability of success of the project under consideration.

6. Exercise 3.2 generalizes this analysis to allow for entrepreneurial risk aversion. Exercise 3.12 considers risk-averse investors.

7. The investors have rate of time preference equal to 0, which is also the market rate of interest.

to switch for a slightly lower interest rate.[8] We use the plural "lenders" even though a single lender may turn out to finance the entire loan, because we want to emphasize that lending is a passive and anonymous activity in the theories reviewed in Part II.

Let us turn to the loan contract. A contract first stipulates whether the project is financed.[9] If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure (the gross payoffs are the *ex post* monetary payoffs and take no account of past investments and private benefit). Intuitively, there is no point in specifying a positive transfer from the lenders to the borrower, as such a transfer can only weaken incentives, while it has no insurance benefit under risk neutrality. This property will be proved more rigorously and is here taken for granted. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders.[10] To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

The zero-profit constraint for the lenders can be written as

$$p_H R_l = I - A,$$

assuming that the loan agreement induces the borrower to behave (which under our assumptions will be the case). The rate of interest ι is given by

$$R_l = (1+\iota)(I-A) \quad \text{or} \quad 1+\iota = 1/p_H.$$

So, unless $p_H = 1$, the nominal rate of interest ι reflects a default premium and exceeds the expected rate of return (called r in Part VI and here normalized to 0) demanded by investors.

We summarize the timing in Figure 3.1.

We assume that the project is viable only in the absence of moral hazard. That is, the project has

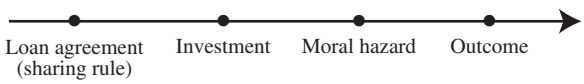


Figure 3.1

positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \tag{3.1}$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \tag{3.2}$$

It is easy to see that inequality (3.2) implies that no loan that gives an incentive to the borrower to misbehave will be granted. Indeed, rewrite (3.2) as

$$[p_L R_l - (I-A)] + [p_L R_b + B - A] < 0.$$

So, in the case of misbehavior, either the lenders must lose money in expectation, or the borrower would be better off using her cash for consumption, or both.

3.2.2 The Lenders' Credit Analysis

Because the project has negative NPV in the case of misbehavior, the loan agreement must be careful to preserve enough of a stake for the borrower in the enterprise. The borrower faces the following tradeoff once the financing has been secured: by misbehaving, she obtains private benefit B, but she reduces the probability of success from p_H to p_L. Because she has stake R_b in the firm's income (she receives R_b in the case of success and 0 in the case of failure), the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \tag{IC$_b$}$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H\left(R - \frac{B}{\Delta p}\right).$$

8. See Exercise 3.13 for the extension of the model to lender market power.

9. "Random financing" contracts, in which the borrower brings equity in exchange for a probability between 0 and 1 of being financed may in some cases be optimal when the investment size is fixed (as it is here) or more generally in the presence of indivisibilities or increasing returns to scale (see Exercise 3.1). For simplicity, we focus on deterministic contracts.

10. The lenders' net payoff is thus $R_l - (I-A)$ in the case of success, and $-(I-A)$ in the case of failure. The borrower's net payoff is thus $R_b - A$ in the case of success, and $-A$ in the case of failure, to which, in both cases, must be added a private benefit B if shirking occurs.

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) \geqslant I - A, \qquad (\mathrm{IR_l})$$

where "$\mathrm{IR_l}$" stands for the lenders' individual rationality constraint (which we will also often call the "breakeven constraint" or the "participation constraint"). Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_{\mathrm{H}}\frac{B}{\Delta p} - (p_{\mathrm{H}}R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \quad \Longleftrightarrow \quad p_{\mathrm{H}}R - I < p_{\mathrm{H}}\frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. Condition (3.4) says that the NPV is smaller than the minimum expected rent that must be left to the borrower to provide her with an incentive to behave.

Thus, *the borrower must have enough assets in order to be granted a loan*. Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. With insufficient assets, the entrepreneur must borrow a large amount and therefore pledge a large fraction of the return in the case of success. The entrepreneur then keeps only a small fraction of the monetary gain and is demotivated. The two parties cannot find a loan agreement that both induces effort (which requires a high compensation for the borrower in the case of success) and allows the lenders to recoup their investment. There is *credit rationing*. A rationed borrower may be willing to give a high fraction of the return to the lenders,[11] which here is equivalent to be willing to pay a high interest rate. But the lenders do not want to grant such a loan.

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_{l} to competitive investors so as not to leave them with a surplus:

$$p_{\mathrm{H}}R_{\mathrm{l}} = I - A.$$

Her stake,

$$R_{\mathrm{b}} = R - R_{\mathrm{l}} = R - \frac{I - A}{p_{\mathrm{H}}} \geqslant R - \frac{I - \bar{A}}{p_{\mathrm{H}}} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

As the conventional wisdom goes, "one only lends to the rich." The threshold $\bar{A}$ has a natural interpretation. As noted earlier, the term $p_{\mathrm{H}}B/\Delta p$ is nothing but the minimum expected monetary payoff to be left to the borrower to preserve incentives; it will be called the *agency rent*. The borrower must make an initial contribution at least equal to $\bar{A}$ so as to reduce the agency rent net of the initial downpayment A to at most the monetary profit $p_{\mathrm{H}}R - I$ of the project.

Using the breakeven condition for the lenders ($p_{\mathrm{H}}R_{\mathrm{l}} = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_{\mathrm{b}} = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_{\mathrm{H}}R_{\mathrm{b}} - A = p_{\mathrm{H}}(R - R_{\mathrm{l}}) - A & \\ \qquad = p_{\mathrm{H}}R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded*.[12]

So, the borrower's utility jumps up at $A = \bar{A}$. While the discontinuity is an artefact of the rigidity of the level of investment, the fact that 1 unit of assets may be worth more than 1 to the borrower in a situation of asymmetric information is quite general. Indeed, in the continuous investment version of this model to be developed in Section 3.4, we will see that for the borrower *assets or net worth have a shadow value exceeding* 1.

Determinants of credit rationing. To sum up, two factors may make a firm credit-constrained in this model:[13]

(i) low amount of cash on hand (low A);

11. This will be the case if A is small.

12. This property holds only in equilibrium. Were the entrepreneur to deviate and misbehave, the entrepreneur's (off-the-equilibrium-path) utility would exceed the smaller (off-the-equilibrium-path) NPV (at least for A close to $\bar{A}$), since the lenders would lose money.

13. The market interest rate, here normalized at 0, is another determinant of the strength of the balance sheet. More generally, the pledgeable income must exceed the investors' outlay times $(1 + r)$,

(ii) high agency cost, where the agency cost can be measured, fixing the project's NPV, $p_H R$, by the combination of the private benefit B and the likelihood ratio $\Delta p / p_H$.

The entrepreneur's ability to borrow is limited by the nonpledgeability of some $(p_H B/\Delta p)$ of the value to investors. Here moral hazard is determined by two factors: the private benefit B that the entrepreneur can enjoy by misbehaving, and the extent to which the verifiable performance reveals such misbehavior. The informativeness of the performance variable regarding effort is defined by the likelihood ratio $(\Delta p/p_H) = (p_H - p_L)/p_H$.[14] This ratio measures the proportional reduction in the probability of success when the entrepreneur misbehaves and is therefore also a measure of the marginal productivity of effort by the borrower. The higher the likelihood ratio, the more informative about effort choice the outcome is ("the better the performance measurement"), and the easier the access to outside financing (in the sense that the minimum net worth $\bar{A}$ decreases). In the model of this section, the pledgeable income never exceeds $p_H R - B$, since the entrepreneur can always take her private benefit B, but may be much smaller when performance measurement is poor, i.e., the likelihood ratio is low.

In practice, the agency cost is influenced not only by the project's and the entrepreneur's characteristics, but also by the surrounding legal, regulatory, and corporate environment. Countries with strong investor protection limit the managers' ability to squander investor money and thereby exhibit lower agency costs; relatedly, the firms' ability to cross-list in jurisdictions with good shareholder protection is expected to reduce their agency cost and therefore to facilitate financing.[15]

Remark (full investment of entrepreneurial assets). We have assumed that the borrower invests her entire wealth. However, it is easy to see that this is an optimal choice for the borrower. Would the borrower want to consume $c \leqslant A$ and invest only $A - c$? If the project is still funded, the borrower still obtains the entire social surplus $p_H R - I$. On the other hand, it becomes more difficult to obtain a loan. Now, the entrepreneur's initial assets must exceed $\bar{A} + c$ in order for the project to be funded. Therefore the entrepreneur cannot gain by not investing her entire wealth in the project.[16]

Remark (high-powered incentive scheme). Earlier we claimed that risk neutrality implies that the absence of reward for the entrepreneur in the case of failure involves no loss of generality. Suppose, more generally, that the entrepreneur receives R_b^S in the case of success and R_b^F in the case of failure, where

$$p_H R_b^S + (1 - p_H) R_b^F \geqslant p_L R_b^S + (1 - p_L) R_b^F + B$$
$$\iff \quad (\Delta p)(R_b^S - R_b^F) \geqslant B$$

in order to discourage the entrepreneur from misbehaving. The investors' income is then

$$p_H(R - R_b^S) + (1 - p_H)(-R_b^F) \leqslant p_H\left(R - \frac{B}{\Delta p}\right) - R_b^F \leqslant \mathcal{P}.$$

Rewarding the entrepreneur in the case of failure implies a uniform upward shift in her minimum incentive-compatible pay structure and an overall reduction in what can be pledged to investors (note the analogy with the previously considered case of an initial consumption c). By contrast, the entrepreneur's utility, provided that she can secure funding, is not affected: because the investors break even, the entire surplus goes to the entrepreneur, who receives

$$U_b = p_H R - I,$$

regardless of the choice of R_b^F. We thus conclude that rewarding the entrepreneur in the case of failure cannot raise her utility, but can compromise financing.

where r is the rate of interest:

$$p_H\left(R - \frac{B}{\Delta p}\right) \geqslant (1 + r)(I - A).$$

Thus, keeping the investment cost I fixed, an increase in the rate of interest r is equivalent to a decrease in the cash on hand.

14. The likelihood ratio is often defined as p_H/p_L. The two notions are obviously equivalent.

15. See, for example, the empirical confirmations by Doidge et al. (2004), Miller (1999), Pagano et al. (2001), and Reese and Weisbach (2002).

16. This reasoning relies, of course, on the borrower's putting equal weight on current and future consumption. If the borrower had immediate consumption needs, she would put some of A aside for consumption. We invite the reader to extend the analysis to the more general specification in which the borrower consumes c_0 at the start and c_1 after the outcome is realized, and has utility the expectation of $u_0(c_0) + u_1(c_1)$, where the functions $u_0(\cdot)$ and $u_1(\cdot)$ are increasing and concave. (The basic insights are unaltered. See also Exercise 3.2.)

Remark (value and investor value). Because the essence of corporate finance is that investors cannot appropriate the full benefit attached to the investments they enable, we must distinguish two slices in the overall cake: that for the insiders and the rest for the outsiders (the decomposition must be finer if there are multiple categories of each). In this book, "value" or "total value" refers to the total cake, while "investor value" refers to the investors' slice; in the barebones model of this section, these two values are pR and pR_l for probability of success p once the investment has been sunk (of course, one needs to subtract I and $I - A$, respectively, if one wants to obtain the corresponding net or *ex ante* magnitudes). The empirical literature often uses the phrase "value" for what we call here "investor value," but this should not create confusion.

Remark (risk taking). Moral hazard here refers to the possibility that the borrower takes an action that reduces investor value (and total value as well). There is no risk taking. We will come back to risk taking in subsequent chapters, but the reader may want to consult Exercises 3.15, 3.16, and 4.15 for three simple ways of introducing risk taking in the context of this simple model.

3.2.3 Do Investors Hold Debt or Equity?

We interpreted the loan agreement as a profit-sharing contract. It turns out that with two levels of profit, 0 and R, the lenders' claim can be thought of as being either debt or equity: put differently, there is here no difference between risky debt and equity. The debt interpretation goes as follows: the borrower must reimburse R_l or else go bankrupt. In the case of reimbursement the borrower keeps the residual $R - R_l$. Alternatively, the two parties can define shares in an all-equity venture. The entrepreneur and the investors hold fractions R_b/R and R_l/R, respectively, of equity. These are called "inside equity" and "outside equity."

This feature of the two-outcome model is both a weakness and a strength. A serious weakness is that it cannot, as it stands, account for the richness of existing securities; but we will show how to extend it in order to generate a more realistic diversity of claims. A strength of this modeling is that it will enable us to analyze a number of key ideas without being held back by the need to specify whether one is analyzing debt, equity, or an alternative claim. Some readers may find it surprising that a lack of predictive power relative to the structure of outside claims may constitute a strength. To clarify this point, it is worth pointing out that many phenomena in corporate finance have wider scope than that defined by the context in which they were discovered. Let us provide some illustrations in support of this view:

(a) As we will study in Chapter 5, Easterbrook (1984) and Jensen (1986) have argued that it is optimal to require cash-rich firms to pay out income on a regular basis, thereby forcing them to return to the capital market. The payment takes the form of a dividend in Easterbrook and of a short-term debt obligation in Jensen. The starting point for both analyses, namely, the desire to pump free cash flow out of the firm, is the same.

(b) The foundations for the soft-budget-constraint problem, also studied in Chapter 5, do not rely on outside claims being debt or equity. While it is usually analyzed in the context of specific assumptions on the financial structure, its logic is quite general.

(c) The literatures on monitoring of a firm by a large shareholder and by a bank holding debt claims have much in common. They are both concerned with the monitor's incentive to supervise and with the impact of monitoring on the firm's behavior.

(d) The idea of using dispersed claimholders to extract rents from third parties (see Chapters 7 and 11) has been developed in separate literatures on debt and on equity.

Thus, abstracting in a first step from the complex issues associated with the diversity of outside claims may generate a better focus on, and a more rigorous analysis of the fundamentals of such phenomena. A richer analysis can then be obtained from the introduction of further modeling features that motivate a diversity of outside claims.

3.2.4 Dilution and Overborrowing

Recall from Section 2.3.3 (see also Fama and Miller 1972) that debt contracts include negative covenants

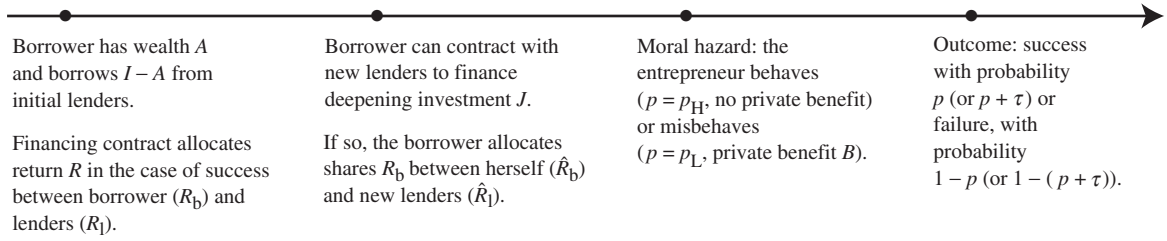


Figure 3.2

prohibiting the dilution of creditors' claims through the issue of new securities, especially ones with equal or higher seniority. There are two basic reasons for such covenants. First, creditors obviously do not want the borrower to issue claims that have a higher or the same seniority as theirs, as this reduces the amount they can collect if the firm defaults. Second, and more subtly, the issue of new securities may alter managerial incentives and the size of the pie.

Let us illustrate the second reason in our simple context. Consider the borrowing contract above in which the lenders take claim R_l in the case of success and the borrower an incentive-compatibility claim $R_b \geqslant B/\Delta p$. Now suppose that there is an opportunity for a "deepening investment." This investment costs an extra J and increases the probability of success uniformly by τ. That is, the probability of success becomes $p_H + \tau$ if the entrepreneur behaves and $p_L + \tau$ if the entrepreneur misbehaves.[17] Assume that this deepening investment is inefficient in that its net cost C_1 is positive, or put differently the expected increase in profit is smaller than J:

$$C_1 \equiv J - \tau R > 0.$$

The timing goes as in Figure 3.2.

We assume away any negative covenant prohibiting further borrowing and so the borrower can contract with new lenders.[18] However, in the case of new financing, initial lenders are not formally diluted in that they keep their stake R_l in success when the borrower contracts with new lenders. So the first motivation for inserting a covenant that prohibits the issuing of new securities is absent.

Note first that it is not in the interest of the borrower to contract with new investors if this results in the same effort, i.e., in no taking of private benefit. Intuitively, the new investment reduces total value by C_1, and so someone must lose in the process. Because the value of the initial investors' claim is increased (to $(p_H+\tau)R_l$) if the borrower still behaves, either the entrepreneur or the new investors must lose, which is impossible because the losing party would refuse to write the second financing contract. So assume that the new financing contract disincentivizes the borrower. This reduced incentive results in a second cost:

$$C_2 \equiv (\Delta p)R - B > 0.$$

As described in the timing, let $\hat{R}_b$ and $\hat{R}_l$ denote the new stake of the borrower and the stake of the new lenders, with

$$\hat{R}_b + \hat{R}_l = R_b.$$

Assuming that the new lenders are competitive, then

$$(p_L + \tau)\hat{R}_l = J.$$

The entrepreneur gains from overborrowing if and only if

$$(p_L + \tau)\hat{R}_b + B > p_H R_b,$$

or, using the breakeven condition for the new investors,

$$[(p_L + \tau)R_b - J] + B > p_H R_b.$$

After some manipulations, this condition becomes

$$[p_H - (p_L + \tau)]R_l > C_1 + C_2.$$

This necessary and sufficient condition for the deepening investment to be financed has a simple

17. This additivity property is convenient because it separates the incentive compatibility constraint from the impact of the new investment.

18. More generally, the division of the pie ($R_l + R_b = R$) is not made contingent on the event of a deepening investment.

interpretation. The right-hand side is the total cost of refinancing: direct cost plus incentive cost. The left-hand side of the inequality is the *externality* on the initial investors. Thus the total cost must be smaller than the loss of value for the initial investors.

When the borrower's balance sheet (as measured by A, say) improves, R_b increases, R_l decreases, and so this inequality is less likely to be satisfied. Put differently, in the absence of negative covenant, *overborrowing is more likely to happen with weak borrowers*.

Let us conclude this analysis of overborrowing with a few remarks. First, overborrowing in this situation can alternatively be avoided by forcing the entrepreneur not to dilute her own claim; this requirement is usually included in compensation contracts, although there have been attempts to evade it through derivative contracts (see Section 1.2.2). Second, the financing contracts need not be signed sequentially: simultaneous contracts also give rise to an overborrowing problem (see Bizer and DeMarzo 1992; Segal 1999). Third, the overborrowing problem arises with a vengeance in the context of sovereign borrowing, in which it is hard to specify a limit on indebtedness of the sovereign, if only because there are many different ways for a government to add new liabilities (see Bolton and Jeanne (2004) for an analysis of sovereign borrowing with the possibility of dilution). Finally, *in a multiperiod financing context, uncoordinated lending further leads to excessively short maturity structures of debt*, as investors scramble to obtain priority over other investors (see Exercise 5.9).

3.2.5 Boosting the Ability to Borrow: Reputational Capital and Capability

Recall from Chapter 2 that lenders do not only look at tangible assets such as cash, land, and equipment. Ceteris paribus, they are more likely to issue a loan if the borrower has a good reputation, as was stressed in particular by Diamond (1991). The role of this intangible capital is easily analyzed in the credit rationing model.

Suppose, for example, that the borrower has less attractive opportunities for misbehavior, in that the private benefit B from misbehaving is reduced to $b < B$.[19] This may have several interpretations. Along the lines of the "effort interpretation" of moral hazard, one might imagine that the project falls well within the core competency of the entrepreneur and therefore demands less attention or supervision of the subordinates: the task is just easier for the entrepreneur. Alternatively, one could imagine that the entrepreneur has less attractive outside options (focusing on other, separate projects of her own) or opportunities for fraud and embezzlement (e.g., it is harder to buy inputs at an inflated price from a friend or family).

With reduced scope for moral hazard, the asset threshold is accordingly lower: from equation (3.3),

$$\bar{A}(b) < \bar{A}(B),$$

where

$$\bar{A}(\beta) \equiv p_H \frac{\beta}{\Delta p} - (p_H R - I),$$

and thus

$$\bar{A}(B) - \bar{A}(b) = \frac{p_H}{\Delta p}(B - b) > 0.$$

In this sense, a "more reliable borrower" (that is, a borrower who has a lower private benefit from misbehaving) is more likely to obtain a loan.

How does this fit with the idea that a good reputation helps raise external finance? Suppose now that the private benefit (B or b) is not directly observed by the lenders, who only have the borrower's track record at their disposal. That is, the lenders know whether the borrower's past projects have been successful or whether past loans have been reimbursed. They use this information to update their beliefs about the reliability of the borrower. A better track record is an (imperfect) indicator of good reliability, that is, in our example, of a low private benefit from misbehaving.

Consider an entrepreneur who got a loan for a first project, and may in the future have new projects that will also call for outside financing. Let us further assume that these future projects are not yet well-defined, and focus on short-term finance. (Chapter 5 will analyze long-term loans.) In this situation, the entrepreneur should adopt a long-term perspective.

19. We could alternatively analyze the impact of a higher probability of success or of changes in other variables, with similar insights. The focus on the private benefit allows a cleaner analysis because changes in the private benefit keep the NPV of the project constant.

That is, she should not content herself with comparing the private benefit and the monetary payoff attached to the first project; she should also take into account the fact that a current success will bring two further benefits:

A *retained-earnings benefit*: even under symmetric information between the parties about the entrepreneur's reliability, a current success helps the entrepreneur build up net worth. This net worth has a shadow value; a unit of income is valued above 1 by the entrepreneur if there is a probability of credit rationing in the future. This benefit is studied in Exercise 3.11.

A *reputational benefit*: if, furthermore, the lenders have incomplete information about the entrepreneur's reliability, their updating of beliefs about this reliability confers an extra benefit on the entrepreneur in the case of success. Reputation complements net worth in reducing the probability of future credit rationing.[20]

An implication of the existence of this reputational benefit is that an unreliable borrower who would have no incentive to behave were her unreliability known to the lenders may have an incentive to behave today in order to get a loan tomorrow. The analysis of the situation becomes more complex once we realize that lenders are unlikely to be fools and understand that unreliable borrowers may have an incentive to masquerade as reliable ones. A proper study of reputational capital requires some (at least intuitive) understanding of dynamic games with incomplete information (see Exercise 6.3). We hope that the idea that reputational capital can substitute for net worth to thwart credit rationing is clear enough. There is indeed empirical evidence that reputation helps borrowers to obtain credit as well as better terms (see, for instance, Banerjee and Duflo's (2000) study of the Indian software industry).

Remark (information sharing). The impact of reputational capital is stronger, the more widely the information about borrower performance is disseminated. Padilla and Pagano (2000) observe that information sharing among lenders reinforces the borrowers' incentives to perform and argue that this may account for the fact that lenders (banks, finance companies, and retailers) spontaneously provide information about past defaults, delays in payment, current debt exposure, and riskiness of their borrowers to credit bureaus and credit-rating agencies, and therefore to their competitors. They develop a model in which lenders may share information even when this may encourage consumer poaching and thus enhanced *ex post* competition.

3.2.6 Making Efficient Use of Information to Reduce the Agency Cost

A basic theoretical result in the economics of agency, due to Holmström (1979), states that making economic agents accountable for events over which they have no control does not help with moral-hazard problems and generally worsens incentives. Roughly speaking, one should try to use the most informative or precise measurement of the agent's economic activity, or what is called in statistics a "summary" or "sufficient statistic."[21] This result underlies much of the thinking about managerial compensation, for example, the quest for good metrics to reward employees (based on customer satisfaction, reduction of unit costs, sales, etc.) or division managers (like EVA (economic value added) or balanced scorecard methods). More to the point for our context, it offers theoretical foundations for the use of benchmarking. Benchmarking, also called relative performance evaluation, consists in comparing the performance of, say, a firm with that of similar firms, to better assess managerial accomplishments. For example, a car producer's good financial performance is less indicative of good management if other car producers also do well than if the automobile industry is in a

20. Things are actually a bit more complicated than this dual benefit suggests: the reputational benefit depends on the borrower's equilibrium behavior (which itself depends on the retained-earnings benefit) and not only on the reputational one. Technically, if the retained-earnings benefit is strong enough to induce a high-private-benefit entrepreneur to behave, then success brings no reputational benefit.

21. A good introduction to sufficient statistics is Chapter 9 of DeGroot (1970). Suppose that one observes two variables x and y, and that one is trying to infer a third, unobservable variable z. The joint distribution of x and y, given z, is $f(x, y \mid z)$. The variable x is a sufficient statistic for (x, y) if the posterior distribution of z conditional on the observation of x and y depends only on x. To recognize sufficient statistics, a necessary and sufficient condition is the factorization criterion, that is, the existence of functions g and h such that $f(x, y \mid z) = g(x, y)h(x, z)$. A simple computation then shows that the distribution of z conditional on x and y does not depend on y.

recession. Or, a high price fetched by the stock of a software or biotechnology start-up company in an initial public offering (IPO) is not foolproof evidence of good entrepreneurship and careful venture capital monitoring if this price is reached during a stock price bubble.

We will come back a few times in this book to the issue of the quality of performance measurement and how it affects the ability to receive financing.[22] Let us just observe that in our context, the ability to raise financing is enhanced by conditioning entrepreneurial compensation on the performance measure with the highest available likelihood ratio.[23]

Let us provide a first illustration of this principle.

Benchmarking. A possible reinterpretation of our model is that there are three states of nature.

(i) Favorable state (probability p_L). The environment is sufficiently favorable that the project will succeed regardless of the entrepreneur's effort.

(ii) Unfavorable state (probability $1 - p_H$). The environment is harsh and the project will fail even if the entrepreneur does her best.

(iii) Intermediate state (probability $\Delta p = p_H - p_L$). Success is not guaranteed, but is reached provided the entrepreneur exerts effort.

Of course, no one *ex ante* knows which state prevails. The financing and effort decisions are chosen in the ignorance of the state of nature.[24]

Suppose now that one will learn *ex post* whether the state was favorable or not (i.e., intermediate or unfavorable), say, by looking at a less promising firm in the same industry that succeeds only if circumstances are favorable. Consider the following compensation scheme:

- the entrepreneur receives 0 if the state is favorable;
- the entrepreneur otherwise receives R_b in the case of success and 0 in the case of failure.

The incentive constraint is still

$$(\Delta p)R_b \geqslant B \qquad (IC_b)$$

since the entrepreneur's stake is still R_b in the state of nature in which she affects profit. The pledgeable income, however, has increased since one no longer pays the entrepreneur for being lucky: now the maximal pledgeable income is

$$p_H R - (\Delta p)\Big[\min_{\{IC_b\}} R_b\Big] = p_H R - B,$$

where $[\min_{\{IC_b\}} R_b]$ denotes the smallest reward R_b that ensures incentive compatibility.

Next, let us assume that the firm's performance can be compared with that of an identical firm facing the same state of nature. Assuming that the entrepreneur in the other firm behaves, then "success" in the other firm provides information that the state is either favorable or intermediate, while "failure" in the other firm reveals an unfavorable state. Then, conditional on the entrepreneur failing, one learns either that she was unlucky or that she failed because she misbehaved. In this case, the pledgeable income cannot be increased by benchmarking if one abides by the entrepreneur's limited liability:[25] when

22. See, for example, Section 4.4 and Chapter 9.

23. For example, it would never come to one's mind to condition the entrepreneur's compensation on the weather in Bali, on the outcome of the soccer World Cup, or on other "irrelevant" variables. Why? Let us be a bit more technical here. In the notation of footnote 21, let (x, y) denote the verifiable state of nature (which includes, but is not limited to, the firm's profit $x \in \{0, R\}$), on which the entrepreneur's reward R_b can be conditioned. Thus, let $R_b(x, y)$ denote the state-contingent compensation specified by the financing contract. Suppose that the firm's profit x is a sufficient statistic for (x, y) when assessing the entrepreneur's effort, which we will here call $z \in \{L, H\}$ (see footnote 21 for the definition of a sufficient statistic). The density of the verifiable state (x, y) for a given effort z can be factorized: $f(x, y \mid z) = g(x, y)h(x, z)$. Thus for a choice of effort $z \in \{L, H\}$, the entrepreneur's expected reward is

$$\int_x \int_y R_b(x, y)g(x, y)h(x, z)\,dx\,dy = \int_x \hat{R}_b(x)h(x, z)\,dx,$$

where $\hat{R}_b(x) \equiv \int_y R_b(x, y)g(x, y)\,dy$.

So, a contract that rewards the entrepreneur solely as a function for profit ($\hat{R}_b(x)$) can do at least as well as a more general contract. And, in general, it can do better (in our context, it does strictly better in particular if $\int_y R_b(0, y)g(0, y) > 0$ and if a strictly positive borrower payoff in the case of success jeopardizes financing). Added risk is bad when the limited liability constraint is binding (and would be bad if the agent were risk averse even if she is not protected by limited liability).

24. Information accrues *ex post* through profit realization. Still the state is not learned *ex post* in the basic model.

25. Benchmarking could become relevant again in this example if we relaxed the limited liability constraint by introducing reputational concerns such as a stigma that affects future borrowing or other future relationships of the borrower or else costly nonmonetary penalties (jail or costly collateral pledging as in Chapter 4). Then, the observation that both entrepreneurs fail implies that the state of nature was unfavorable and so stigmas and/or nonmonetary penalties are not in order, unlike the situation in which only one entrepreneur fails (which implies that she misbehaved).

the entrepreneur fails, she already receives 0. On the other hand, it would be optimal to punish the entrepreneur harshly when she fails and the benchmark firm succeeds.[26]

3.2.7 Sensitivity of Investment to Cash Flow: A First Look

Recall from Section 2.5 the empirical finding that investment is sensitive to cash flow. An interesting issue is whether this "investment–cash flow sensitivity" increases with the extent to which the firm is financially constrained. Fazzari et al. (1988) use *a priori* measures of financial constraints and find that the sensitivity of investment to cash flow is particularly large for firms that have trouble raising external funds (for example, firms facing high agency costs). Kaplan and Zingales (1997) argue that there is no theoretical basis for this relationship and present empirical evidence that differs from that of Fazzari et al.

Although the model in this chapter is static while the empirical evidence relates to ongoing concerns (multistage financing is studied in Chapter 5), it can shed some light on the debate. We can imagine that cash on hand A includes the cash flow accruing from the firm's previous activity and see how investment reacts to a small change in the cash flow.[27]

There is a sense in which Fazzari et al. (1988) are right on the theoretical front: the firms whose investment is boosted by a small increase in cash flow are the marginal firms, i.e., those whose cash on hand A lies just below $\bar{A} = I - \rho_0$, where ρ_0 denotes the pledgeable income:

$$\rho_0 \equiv p_H\left(R - \frac{B}{\Delta p}\right) = p_H R - \frac{p_H B}{\Delta p}.$$

Firms with more cash or a lower agency cost do not modify their investment behavior as their investment was already unconstrained.

Suppose, however, that firms are heterogeneous in the two dimensions: cash A and pledgeable income ρ_0 (we normalize the investment I to be the same for all). Assume for simplicity that these two variables are independently distributed (there is no reason for this to be the case: for example, firms with higher pledgeable income may have been able to invest more in the past and be richer today). Let $G(A)$ denote the (continuous) cumulative distribution of cash among firms in the economy, with density $g(A)$. Because only firms with cash on hand A satisfying $\rho_0 \geqslant I - A$ receive financing, aggregate investment among firms with pledgeable income ρ_0 is

$$\mathcal{I}(\rho_0) \equiv [1 - G(I - \rho_0)]I.$$

Now, consider a small, uniform increase in cash δA for all firms. Then, investment among firms characterized by ρ_0 increases by

$$\delta\mathcal{I}(\rho_0) = g(I - \rho_0)I\delta A.$$

And so

$$\frac{\partial}{\partial \rho_0}(\delta\mathcal{I}(\rho_0)) = -g'(I - \rho_0)I\delta A.$$

If the density is decreasing ($g' < 0$), the sensitivity of investment to cash flow is higher for firms with a *low* agency cost (a high ρ_0) as in Kaplan and Zingales; intuitively, the cutoff $\bar{A}$ for firms with a low agency cost is low, and so with a decreasing density there are a lot of marginal firms. With an increasing density ($g' > 0$), the sensitivity of investment to cash flow is higher for firms with a *high* agency cost (a low ρ_0), as in Fazzari et al. Thus, unless one has more precise information about the actual heterogeneity of firms, it is difficult to predict how the sensitivity of investment to cash flow varies with an *a priori* measure of financial constraints (a proxy for (minus) ρ_0).

26. This is one aspect in which a "limited liability model" differs from a "risk-aversion model." In the rest of Section 3.2, we might as well have assumed that the entrepreneur is very risk averse at her subsistence level, normalized at zero consumption, that is, her utility falls very quickly at that level. Suppose at the extreme that the entrepreneur gets $-\infty$ when receiving a negative income. Then, provided that $p_H < 1$ (the entrepreneur may behave and be unlucky), it would not be optimal to set rewards below the subsistence level.

27. This thought experiment in a sense consists in looking at a single period of an ongoing firm that engages in short-term borrowing from investors. There are two reasons why this is only a first step toward an understanding of the sensitivity of investment to cash flow. First, if the firm anticipates that it may be credit-constrained tomorrow, the shadow value of 1 unit of profit at the end of the period exceeds 1, as it may help overcome financing problems in the future. More importantly, this shadow value may vary with current investment. Second, the description of the financial arrangements as a sequence of short-term borrowing contracts misses important long-term financing features (credit lines, debt–equity ratio, maturity structure of debt, etc.) that have an important impact on financial constraints (see Chapter 5).

3.3 Debt Overhang

Following Myers (1977), a number of contributions have studied situations in which a borrower is debt-ridden and unable to raise funds for an otherwise profitable project. The borrower is then said to suffer from debt overhang. The framework just developed suggests two possible interpretations of debt overhang. The first interpretation pursued below is a mere reinterpretation of the credit rationing analysis above: previous investors' collateral claim on the firm's assets reduces the net worth to below the threshold asset level for financing the new investment. Furthermore, the new project overall produces too little pledgeable income and so investment does not take place even if previous investors are willing to renegotiate their claim. The second and more interesting interpretation, and that stressed by the literature, emphasizes the need for renegotiating past liabilities in order to enable new investments.

3.3.1 Decrease in Net Worth

First, the borrower may have a positive-NPV project that would be financed in the absence of any previous debt obligation, but is denied financing due to such an obligation. Namely, suppose that (i) the entrepreneur has A in cash or collateral, but owes D from previous borrowing to a group of investors whom we will call the "initial investors," (ii) the initial investors have insisted on a covenant specifying that the borrower cannot raise more funds without their consent, and (iii) the borrower's assets A are pledged to the initial investors as collateral in case of default. If[28]

$$A > \bar{A} > A - D \geqslant 0,$$

the project would have been financed in the absence of previous borrowing but is not undertaken, since investors as a whole, that is, the initial investors and new investors (who can, of course, be the initial investors themselves), cannot recoup the cost of their investment $(I - A)$ plus the previous debt obligation (D) while they can receive D by seizing the collateral. More precisely, suppose the borrowers, the initial investors, and the new investors enter an agreement so as to finance the project. Because initial investors can secure themselves D by seizing the collateral, they must receive an expected payment at least equal to D under this agreement. Because the pledgeable income net of the investment cost is equal to

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) - I,$$

new investors obtain at most

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) - I - D + A = A - D - \bar{A} < 0,$$

which contradicts the fact that rational investors must at least break even.

3.3.2 Lack of Renegotiation[29]

Second, and more interestingly, suppose that

(i) the project is sufficiently profitable to attract funds even if the borrower has zero net worth,

$$\bar{A} < 0;$$

(ii) the borrower has previously been granted a long-term loan and is due to reimburse D "at the end," that is, when the outcome of the project (if financed) occurs;

(iii) this long-term debt obligation is contractually *senior* to any claim that the borrower might issue (a senior claim is a claim that must be paid before the borrower or any other claimholder receives any money);

(iv) the borrower has no cash ($A = 0$); and

(v) the debt overhang problem is sufficiently serious as not to be overcome by the expected profitability of the new project, or, put differently, the "slack" in pledgeable income, $-\bar{A}$, is smaller than what has to be paid back to previous investors, $p_{\mathrm{H}}D$, if the project is funded:

$$\bar{A} + p_{\mathrm{H}}D > 0.$$

28. Recall that

$$\bar{A} = p_{\mathrm{H}}\frac{B}{\Delta p} - (p_{\mathrm{H}}R - I)$$

is the minimum net worth to obtain financing.

29. The notion that renegotiation breakdowns generate debt overhang is central to Myers's (1977) original analysis, and also underlies that in Hart and Moore (1995) and Bhattacharya and Faure-Grimaud (2001). We will describe the debt overhang situation as one in which a new investment cannot be financed solely because renegotiation with previous debtholders proves infeasible. Debt overhang is generally described in the literature as a situation in which a firm may not be able to continue because it cannot renegotiate with its creditors. It is clear that the two situations are formally equivalent. The act of spending money to let a distressed firm continue is equivalent to an investment.

Because the borrower has no cash, initial investors receive nothing if the project is not financed. So, they are willing to participate in the financing of the project as long as they break even on this investment. For example, they can forgive existing debt, finance the investment I, and demand the entire cash-flow rights attached to external shares, that is, $R - B/\Delta p$ in the case of success. The initial investors then obtain

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) - I = -\bar{A} > 0.$$

The borrower is willing to go along with this arrangement, which allows her to continue and obtain rent $p_{\mathrm{H}}B/\Delta p$ in expectation rather than 0 if the project is not financed.

Suppose next that the initial investors have no cash and thus cannot directly finance the investment I. The borrower then needs to turn to new investors. Are the latter willing to finance the project? Because the initial debt is senior, and because the borrower needs to keep a minimum stake in the firm in order to commit to behave, at most

$$R - \frac{B}{\Delta p} - D$$

can be pledged to new investors in the case of success (and 0 in the case of failure). New investors are willing to enter an agreement to finance the project if and only if

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p} - D\right) \geqslant I$$

or

$$\bar{A} + p_{\mathrm{H}}D \leqslant 0,$$

which contradicts assumption (v).

To sum up, *the borrower cannot raise funds from new investors if she does not renegotiate some debt forgiveness from initial investors*. If renegotiation with initial investors is infeasible, gains from trade between the borrower and the community of investors may not be realized. Renegotiation breakdown creates debt overhang.

The possibility of debt overhang is often invoked in contexts in which "initial investors" stand for "corporate bondholders." It is often thought that because they are dispersed, and despite the existence of some coordinating mechanisms (nomination of a bond trustee, possibility for the firm to offer new securities in exchange for the bonds), bondholders have trouble renegotiating their claim when the borrower faces distress and requires some debt forgiveness.

In contrast, let us assume that initial investors are able to act collectively and renegotiate their initial claims. Because $\bar{A} < 0$, we know that there exists some renegotiated arrangement that is agreeable to all parties (borrower, initial investors, new investors), who would all get nothing if they failed to reach an agreement. Suppose, for example, that the initial investors accept a reduction in the face value of the debt from D to $d < D$, where

$$\bar{A} + p_{\mathrm{H}}d = 0.$$

Then new investors receive

$$\left(R - \frac{B}{\Delta p} - d\right)$$

in the case of success and are therefore willing to invest, since

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p} - d\right) = I$$

is equivalent to their breakeven constraint (3.3):

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) = I - \bar{A}.$$

Initial investors benefit from forgiving some of their claim as they now get

$$p_{\mathrm{H}}d = -\bar{A} > 0.$$

Lastly, the borrower can undertake the project and obtains rent $p_{\mathrm{H}}B/\Delta p > 0$.

Debt renegotiation thus allows the project to be undertaken and all parties to share the resulting gains from trade. How these gains from trade are actually shared depends, of course, on the relative bargaining power of the borrower and the initial investors (the new investors being assumed to be competitive and thus to just break even). The arrangement described above corresponds to the renegotiation that is most favorable to the initial investors. But, by varying continuously the relative bargaining power of the borrower and initial investors, one can generate any level of debt forgiveness from $D - d$ (the most favorable to initial investors) to D (the least favorable to them).

3.4 Borrowing Capacity: The Equity Multiplier

3.4.1 The Continuous-Investment Model

The continuous-investment model of this section is the polar opposite of the fixed-investment one. The fixed-investment model depicts a situation in which returns are sharply decreasing beyond some investment level. In contrast, we now assume that there are constant returns to scale in the investment technology. An investment $I \in [0, \infty)$ yields income RI, proportional to I, in the case of success, and 0 in the case of failure. The borrower's private benefit from misbehaving is also taken to be proportional to investment. As before, the borrower has a choice between behaving, in which case she derives no private benefit and the probability of success is p_H, and misbehaving, that is, enjoying private benefit BI, and reducing the probability of success to $p_L = p_H - \Delta p < p_H$. (One can also analyze the intermediate case of a continuous investment with general decreasing returns to scale (see Exercise 3.5).)

The borrower initially has cash A, and must therefore borrow $I - A$ to finance a project of size I. A loan agreement specifies that the lenders (who as before are assumed to make no profit) and the borrower receive 0 each in the case of failure, and R_l and R_b, respectively, in the case of success, where $R_l + R_b = RI$.

As in Section 3.2, we assume that investment has positive NPV (net present value), here per unit of investment, if the borrower behaves,

$$p_H R > 1, \tag{3.7}$$

but negative NPV otherwise,

$$1 > p_L R + B, \tag{3.8}$$

so that unless one can control the agency problem the investment cannot be funded. We also make an assumption that guarantees that the equilibrium investment is finite:

$$p_H R < 1 + \frac{p_H B}{\Delta p}. \tag{3.9}$$

Like inequality (3.5) in Section 3.2, inequality (3.9) has a simple interpretation: the expected net revenue per unit of investment, $p_H R - 1$, is lower than the per-unit agency cost, $p_H B / \Delta p$.

Finally, we keep assuming that the capital market is competitive. The analysis is very similar when the borrower faces a lender with market power, except that the resulting investment scale is smaller (see Exercise 3.13).

3.4.2 The Lenders' Credit Analysis

Following the steps of Section 3.2, the incentive compatibility and the breakeven conditions are

$$(\Delta p) R_b \geqslant BI \tag{IC$_b$}$$

and

$$p_H (RI - R_b) \geqslant I - A. \tag{IR$_l$}$$

In equilibrium, competitive lenders make no profit on the contract that is most advantageous for the borrower; the borrower's net utility is therefore equal to the social surplus brought about by the investment:

$$U_b = (p_H R - 1) I. \tag{3.10}$$

From (3.10) it is optimal for the borrower to invest as much as possible. The upper bound on investment and in turn her borrowing capacity ("outside financing capacity" or "debt capacity") are determined by constraints (IC$_b$) and (IR$_l$). Substituting (IC$_b$) into (IR$_l$), we obtain

$$I \leqslant kA, \tag{3.11}$$

where

$$k = \frac{1}{1 - p_H (R - B/\Delta p)} > 1. \tag{3.12}$$

The denominator of k is positive from (3.9). Furthermore, conditions (3.7) and (3.8) imply that $(\Delta p) R > B$, and therefore that the denominator of k is smaller than 1. This is important: the fact that $k > 1$ shows that the borrower can lever her wealth, k being the multiplier.

The multiplier is smaller, the higher the private benefit (B) and the lower the likelihood ratio ($\Delta p / p_H$, fixing p_H and thus the profitability of the investment), which are our two measures of the agency cost.

Conditions (3.7) and (3.10) furthermore imply that it is optimal for the borrower to invest k times her cash A, that is, to borrow $d = (k - 1)$ times her level of cash, where

$$d = \frac{p_H (R - B/\Delta p)}{1 - p_H (R - B/\Delta p)}. \tag{3.13}$$

The maximum loan, dA, is called "*borrowing capacity.*"[30]

Another important concept (which will be used, for example, in computing the value of retained earnings in a dynamic context) is the *shadow value v of equity (here cash).* The entrepreneur derives gross utility $v > 1$ from one more unit of equity. Letting $U_b^g \equiv A + U_b$ denote the borrower's gross utility, and using (3.10) and (3.11), we have

$$U_b^g \equiv vA, \tag{3.14}$$

where the shadow value of equity is

$$v = \frac{p_H B/\Delta p}{1 - p_H(R - B/\Delta p)} > 1. \tag{3.15}$$

As one would expect (in the relevant range defined by (3.7)–(3.9)), the borrowing capacity increases with per-unit income R and decreases with the extent of the moral-hazard problem (measured by the borrower's private benefit or the inverse of the likelihood ratio). The shadow value of equity increases with per-unit income R and also with the extent of the moral-hazard problem.[31]

Finally, let us introduce some notation that will be used repeatedly throughout the book. Let

$$\rho_1 \equiv p_H R$$

denote the expected payoff per unit of investment and

$$\rho_0 \equiv p_H\left(R - \frac{B}{\Delta p}\right)$$

denote the expected pledgeable income per unit of investment. Assumptions (3.7) and (3.9) can be rewritten as

$$\rho_1 > 1 > \rho_0.$$

The equity multiplier is then

$$k = \frac{1}{1 - \rho_0}, \tag{3.11′}$$

the debt capacity per unit of net worth

$$d = \frac{\rho_0}{1 - \rho_0}, \tag{3.12′}$$

and the borrower's gross utility

$$U_b^g = vA = \frac{\rho_1 - \rho_0}{1 - \rho_0}A. \tag{3.14′}$$

The borrower's net utility can then be written as

$$U_b^n = U_b = U_b^g - A = \frac{\rho_1 - 1}{1 - \rho_0}A = (\rho_1 - 1)I,$$

as one could have expected.

Remark (factors that keep the investment bounded). Condition (3.9) (the condition that the pledgeable income per unit of investment is smaller than 1) was needed in order to keep the investment finite in this constant return to scale environment. Such a condition is no longer needed if the price of output and therefore the revenue in the case of success is not fixed but rather depends on, say, industry investment. An increase in per-firm investment then lowers the market price, reducing both value and pledgeable income (see Exercise 3.17 for more detail).

Remark (sensitivity of investment to cash flow). Let us briefly return to the sensitivity of investment to cash flow. In the variable-investment model,

$$\frac{\partial}{\partial \rho_0}\left(\frac{\partial I}{\partial A}\right) = \frac{1}{(1 - \rho_0)^2} > 0,$$

and so firms with a low agency cost, which are therefore less financially constrained, exhibit a higher sensitivity. Intuitively, such firms have a high multiplier and their investment is therefore more sensitive to available cash.

3.4.3 Collateral Values: Outside Debt and the Maximal Incentives Principle

We now return to the indeterminacy of the financial structure (debt or equity) discussed earlier. It turns out that this indeterminacy was an artefact of the absence of profit in the case of failure.[32]

Thus, assume that, for investment size I, the profit is $R^S I$ in the case of success and $R^F I$ in the case of failure, where R^F is now positive. $R^F I$ can be thought of as the salvage value of assets and

$$RI \equiv (R^S - R^F)I$$

30. Note also that the "gearing ratio" $g = d/k = p_H R - p_H B/\Delta p$ is less than 1, and that the debt-over-inside-equity ratio is equal to d.

31. The shadow value is here constant with wealth A. With a decreasing-returns-to-scale technology, v depends on wealth and $v'(A) < 0$: the marginal wealth enables less and less profitable marginal investments as wealth increases (see Exercise 3.5).

32. The clearest illustration of this point is for the variable-investment model, which is why we treat this here. The same point can be made in a slightly different form (as some indeterminacy may remain) in the fixed-investment version (see Exercise 3.18).

as the increase in profit brought about by success. One would expect R^F to be larger when secondary asset markets are liquid.[33]

The model is otherwise the same as in the rest of Section 3.4: the private benefit (BI in the case of misbehavior, 0 otherwise) is also proportional to investment.

The generalization of the condition that the NPV per unit of investment is positive while the pledgeable income per unit of investment is negative ($p_H R > 1 > p_H(R - B/\Delta p)$) is

$$p_H R + R^F > 1 > p_H\left(R - \frac{B}{\Delta p}\right) + R^F.$$

A contract specifies an investment level I and a sharing rule, or equivalently a reward for the entrepreneur for each performance level: $\{R_b^S, R_b^F\}$, with $R_b^S, R_b^F \geqslant 0$ due to limited liability.

The optimal contract maximizes the entrepreneur's expected compensation,

$$U_b = \max_{\{R_b^S, R_b^F, I\}} \{p_H R_b^S + (1 - p_H) R_b^F - A\},$$

subject to two constraints (that will turn out to be binding at the optimum): the entrepreneur's incentive constraint,

$$(\Delta p)(R_b^S - R_b^F) \geqslant BI,$$

and the investors' breakeven constraint,

$$p_H(R^S I - R_b^S) + (1 - p_H)(R^F I - R_b^F) \geqslant I - A.$$

To show that the investors' breakeven constraint is binding, note that, if it were not, then the entrepreneur could increase R_b^S and R_b^F by an equal and small amount without affecting the incentive compatibility constraint. This uniform increase in compensation would raise the entrepreneur's payoff. As is now familiar, we conclude that the investors receive no surplus, and so (by substituting the breakeven constraint into the objective function) the entrepreneur's utility is equal to the NPV:

$$U_b = (p_H R + R^F - 1) I.$$

Because the NPV per unit of investment is positive, the entrepreneur therefore chooses the highest possible investment.

Next, note that the incentive constraint is binding (otherwise, the optimal investment would be infinite, which would violate the two constraints combined).

Lastly, suppose that $R_b^F > 0$ at the optimum. And consider a small increase $\delta R_b^S > 0$ in managerial compensation in the case of success together with a small decrease $\delta R_b^F < 0$ in the case of failure that keeps the investors' profitability constant:

$$p_H \delta R_b^S + (1 - p_H)\delta R_b^F = 0.$$

This small change (which is feasible only if $R_b^F > 0$) also keeps the objective function constant. But the incentive constraint is now slack, a contradiction. We thus conclude that at the optimum

$$R_b^F = 0.$$

Hence, an all-equity firm cannot be optimal: in the absence of debt, the entrepreneur would receive $R^F I$ times her share of stocks in the firm, and therefore would be rewarded even in the case of failure. By contrast, investors' holding debt $D \geqslant R^F I$ is an optimal financial structure. Using the fact that the two constraints are binding, the borrowing capacity is given by

$$R^F I + p_H\left(R - \frac{B}{\Delta p}\right) I = I - A$$

or

$$I = \frac{A}{1 - [p_H(R - B/\Delta p) + R^F]}. \qquad (3.16)$$

Predictions. The variable-investment model of this section is, of course, much too simplistic to provide even a stylized account of capital structure and investment. It, however, delivers three interesting preliminary insights.

• *Firms with lower agency costs borrow more.* As in Section 3.2.2, the firm's outside financing capacity is higher, the lower the agency cost as measured either by the private benefit B or by (the inverse of) the likelihood ratio $\Delta p/p_H$ (keeping p_H and therefore profitability constant).

• *The investors' holding safe debt plus some equity maximizes the entrepreneur's stake in the project and thereby her incentives.* (We will investigate the generality of this insight in Section 3.5.)

Decomposing the investors' claim into safe debt (which repays $R^F I$) and risky equity (which repays in

33. Several chapters in this book (primarily Chapter 14) will investigate the determinants of asset prices in secondary markets.

expectation $p_H[R - B/\Delta p]I$), the leverage ratios,

$$\frac{\text{debt}}{\text{total equity}} = \frac{R^F I}{p_H R I} = \frac{R^F}{p_H R}$$

and

$$\frac{\text{debt}}{\text{outside equity}} = \frac{R^F I}{p_H(R - B/\Delta p)I} = \frac{R^F}{p_H(R - B/\Delta p)},$$

are both constant in this simple-minded model.

• *Credit rationing is more binding for firms with less tangible assets or assets that have a lower value in liquidation* (there is indeed substantial evidence in this direction: see Chapter 2). To see this, let us decrease the value of tangible assets while keeping the NPV per unit of investment constant: that is, keeping other parameters constant, let us consider a decrease from R^F to $\hat{R}^F$ ($\hat{R}^F < R^F$) of the per-unit salvage value and an increase from p_H to $p_H + \tau$ ($\tau > 0$) of the probability of success such that

$$\hat{R}^F + (p_H + \tau)R = R^F + p_H R.$$

In order to keep the agency problem invariant (so as not to interfere with the first prediction), let us assume that the probability of success in the case of misbehavior becomes $p_L + \tau$. The borrower's incentive compatibility constraint is then unchanged as

$$[(p_H + \tau) - (p_L + \tau)](R_b^S - R_b^F) = (\Delta p)(R_b^S - R_b^F).$$

The analysis is unchanged and the new investment becomes

$$\hat{I} = \frac{A}{1 - [(p_H + \tau)(R - B/\Delta p) + \hat{R}^F]} < I.$$

Thus, ceteris paribus, tangible assets facilitate financing.[34]

3.4.4 Going Forward

This chapter offered a first glance at the basic conflict between value and pledgeable income. When pressed to produce returns to attract investors, borrowers first offer them a large debt repayment or a higher share of profits (Section 3.2). This policy is, however, limited by entrepreneurial moral hazard and must be supplemented by costly "concessions." Technically, with competitive investors, the total value (NPV) goes to the entrepreneur, who aims at maximizing this value subject to the constraint that the pledgeable income be sufficient to enable the investors to recoup their investments. The resulting policy (charter, covenants, governance structure, etc.) therefore sacrifices value to generate enough pledgeable income if the breakeven constraint condition is binding.

The variable-investment model of Section 3.4 pointed at such an elementary concession: a limited investment size. Indeed, with constant returns to scale, it would be optimal for the firm to grow without bounds, but pleasing investors requires a limited size (all the more so, as we have seen, as the agency problem is important and as assets are intangible). The rest of the book will provide further illustrations of the idea that entrepreneurs must sometimes "bend over backwards" in order to attract investors: costly collateral pledging, restricted exit options, short maturity structures, enlisting of active and speculative monitors, allocations of control rights to equityholders and debtholders, limits on takeover defenses, and so forth.

34. Rather than increase the probability of success, we could have increased the payoff R in the case of success. Then investment would have been invariant (see equation (3.16)). As increases in the value of the risky component are in general associated with both types of changes, the conclusion that tangible assets facilitate financing is robust.

Supplementary Sections

3.5 Related Models of Credit Rationing: Inside Equity and Outside Debt

This supplementary section reviews three classic, alternative models of credit rationing. These models are a bit more complex than the basic credit rationing model developed in this chapter and this supplementary section is accordingly more technical than the text. They are not relegated to the supplementary section because they are deemed "less important." Rather, the reader should recall from the introduction that we want to conduct controlled experiments throughout the book. Using the same simple and tractable model throughout allows us to concentrate on the key insights of the theory without getting bogged down by extraneous modeling changes. This is the motivation for setting these models aside. It should furthermore be borne in mind that these

thereof) must be self-sustaining rather than contracted upon. The basic mechanism is otherwise similar to the Bolton–Scharfstein mechanism, in that lenders cannot appropriate any of the current return and count solely on the nonrefinancing threat to recoup their investment. Bulow and Rogoff consider an infinite-horizon, symmetric-information model in which (a) the sovereign can decide not to reimburse and (b) the sovereign can save, and (c) the rate of growth of the economy is smaller than the rate of interest. They show that no lending is feasible as the borrower always prefers to default (and save some of the concomitant extra income).

Several contributions have shown that borrowing may be feasible in more general no-sanction environments. First, Hellwig and Lorenzoni (2004) show that when the rate of growth in the absence of sovereign borrowing exceeds the rate of interest, then sovereign debt borrowing is feasible, even though incentive-compatible repayments still require borrowing levels below the first-best level. Intuitively, exclusion from borrowing is a stronger threat when the rate of growth is large relative to the rate of interest. Second, an outright exclusion, in which the defaulting sovereign cannot even save, makes it particularly costly for the sovereign to repudiate its debt. Again, some sovereign debt may then be issued in equilibrium (Kehoe and Levine 1993; Kocherlakota 1996). Finally, standard "type-based" reputation models (see, for example, Kreps et al. 1982) would deliver some equilibrium borrowing.

3.9 Exercises

Exercise 3.1 (random financing). Consider the fixed-investment model of Section 3.2. We know that if $A \geqslant \bar{A}$, where

$$I - \bar{A} = p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right),$$

it is both optimal and feasible for the borrower to sign a contract in which the project is undertaken for certain. We also noted that for $A < \bar{A}$, the borrower cannot convince investors to undertake the project with probability 1. With $A > 0$, the entrepreneur benefits from signing a "random financing contract," though.

(i) Consider a contract in which the borrower invests $\hat{A} \in [0, A]$ of her own money, the project is financed with probability x, and the borrower receives R_{b} in the case of success and 0 otherwise. Write the investors' breakeven condition.

(ii) Show that (provided the NPV, $p_{\mathrm{H}}R - I$, is positive) it is optimal for the borrower to invest

$$\hat{A} = A.$$

How does the probability that the project is undertaken vary with A?

Exercise 3.2 (impact of entrepreneurial risk aversion). Consider the fixed-investment model developed in this chapter: an entrepreneur has cash amount A and wants to invest $I > A$ into a project. The project yields $R > 0$ with probability p and 0 with probability $1 - p$. The probability of success is p_{H} if the entrepreneur works and $p_{\mathrm{L}} = p_{\mathrm{H}} - \Delta p$ ($\Delta p > 0$) if she shirks. The entrepreneur obtains private benefit B if she shirks and 0 otherwise. Assume that

$$I > p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right).$$

(Suppose that $p_{\mathrm{L}}R + B < I$; so the project is not financed if the entrepreneur shirks.)

(i) In contrast with the risk-neutrality assumption of this chapter, assume that the entrepreneur has utility for consumption c:

$$u(c) = \begin{cases} c & \text{if } c \geqslant c_0, \\ -\infty & \text{otherwise.} \end{cases}$$

(Assume that $A \geqslant c_0$ to ensure that the entrepreneur is not in the "$-\infty$ range" in the absence of financing.)

Compute the minimum equity level $\bar{A}$ for which the project is financed by risk-neutral investors when the market rate of interest is 0. Discuss the difference between $p_{\mathrm{H}} = 1$ and $p_{\mathrm{H}} < 1$.

(ii) Generalize the analysis to risk aversion. Let $u(c)$ denote the entrepreneur's utility from consumption with $u' > 0$, $u'' < 0$. Conduct the analysis assuming either limited liability or the absence of limited liability.

Exercise 3.3 (random private benefits). Consider the variable-investment model: an entrepreneur ini-

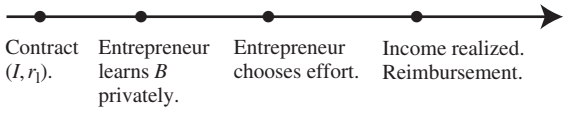


Figure 3.10

tially has cash A. For investment I, the project yields RI in the case of success and 0 in the case of failure.

The probability of success is equal to $p_H \in (0,1)$ if the entrepreneur works and $p_L = 0$ if the entrepreneur shirks. The entrepreneur obtains private benefit BI when shirking and 0 when working. The per-unit private benefit B is unknown to all *ex ante* and is drawn from (common knowledge) uniform distribution F:

$$\Pr(B < \hat{B}) = F(\hat{B}) = \hat{B}/R \quad \text{for} \quad \hat{B} \leqslant R,$$

with density $f(\hat{B}) = 1/R$. The entrepreneur borrows $I - A$ and pays back $R_1 = r_1 I$ in the case of success. The timing is described in Figure 3.10.

(i) For a given contract (I, r_1), what is the threshold B^*, i.e., the value of the private per-unit benefit above which the entrepreneur shirks?

(ii) For a given B^* (or equivalently r_1, which determines B^*), what is the debt capacity? For which value of B^* (or r_1) is this debt capacity highest?

(iii) Determine the entrepreneur's expected utility for a given B^*. Show that the contract that is optimal for the entrepreneur (subject to the investors breaking even) satisfies

$$\tfrac{1}{2} p_H R < B^* < p_H R.$$

Interpret this result.

(iv) Suppose now that the private benefit B is observable and verifiable. Determine the optimal contract between the entrepreneur and the investors (note that the reimbursement can now be made contingent on the level of private benefits: $R_1 = r_1(B)I$).

Exercise 3.4 (product-market competition and financing). Two firms, $i = 1, 2$, compete for a new market. To enter the market, a firm must develop a new technology. It must invest (a fixed amount) I. Each firm is run by an entrepreneur. Entrepreneur i has initial cash $A_i < I$. The entrepreneurs must borrow from investors at expected rate of interest 0. As in the single-firm model, an entrepreneur enjoys private benefit B from shirking and 0 when working. The probability of success is p_H and $p_L = p_H - \Delta p$ when working and shirking.

The return for a firm is

$$R = \begin{cases} D & \text{if both firms succeed in developing the technology (which results in a duopoly),} \\ M & \text{if only this firm succeeds (and therefore enjoys a monopoly situation),} \\ 0 & \text{if the firm fails,} \end{cases}$$

where $M > D > 0$.

Assume that $p_H(M - B/\Delta p) < I$. We look for a Nash equilibrium in contracts (when an entrepreneur negotiates with investors, both parties correctly anticipate whether the other entrepreneur obtains funding). In a first step, assume that the two firms' projects or research technologies are independent, so that nothing is learned from the success or failure of the other firm concerning the behavior of the borrower.

(i) Show that there is a cutoff $\underline{A}$ such that if $A_i < \underline{A}$, entrepreneur i obtains no funding.

(ii) Show that there is a cutoff $\bar{A}$ such that if $A_i > \bar{A}$ for $i = 1, 2$, both firms receive funding.

(iii) Show that if $\underline{A} < A_i < \bar{A}$ for $i = 1, 2$, then there exist two (pure-strategy) equilibria.

(iv) The previous questions have shown that when investment projects are independent, product-market competition makes it *more difficult* for an entrepreneur to obtain financing. Let us now show that when projects are correlated, product-market competition may *facilitate financing* by allowing financiers to benchmark the entrepreneur's performance on that of competing firms.

Let us change the entrepreneur's preferences slightly:

$$u(c) = \begin{cases} c & \text{if } c \geqslant c_0, \\ -\infty & \text{otherwise.} \end{cases}$$

That is, the entrepreneur is infinitely risk averse below c_0 (this assumption is stronger than needed, but it simplifies the computations).

Suppose, first, that only one firm can invest. Show that the necessary and sufficient condition for investment to take place is

$$p_H\left(M - \frac{B}{\Delta p}\right) - c_0 \geqslant I - A.$$

(v) Continuing on from question (iv), suppose now that there are two firms and that their technologies are *perfectly correlated* in that if both invest and both entrepreneurs work, then they both succeed or both fail. (For the technically oriented reader, there exists an underlying state variable ω distributed uniformly on $[0,1]$ and common to both firms such that a firm always succeeds if $\omega < p_L$, always fails if $\omega > p_H$, and succeeds if and only if the entrepreneur works when $p_L < \omega < p_H$.)

Show that if

$$p_H D - c_0 \geqslant I - A,$$

then it is an equilibrium for both entrepreneurs to receive finance. Conclude that product-market competition may facilitate financing.

Exercise 3.5 (continuous investment and decreasing returns to scale). Consider the continuous-investment model, with one modification: investment I yields return $R(I)$ in the case of success, and 0 in the case of failure, where $R' > 0$, $R'' < 0$, $R'(0) > 1/p_H$, $R'(\infty) < 1/p_H$. The rest of the model is unchanged. (The entrepreneur starts with cash A. The probability of success is p_H if the entrepreneur behaves and $p_L = p_H - \Delta p$ if she misbehaves. The entrepreneur obtains private benefit BI if she misbehaves and 0 otherwise. Only the final outcome is observable.) Let I^* denote the level of investment that maximizes total surplus: $p_H R'(I^*) = 1$.

(i) How does investment $I(A)$ vary with assets?

(ii) How does the shadow value v of assets (the derivative of the borrower's gross utility with respect to assets) vary with the level of assets?

Exercise 3.6 (renegotiation and debt forgiveness). When computing the multiplier k (given by equation (3.12)), we have assumed that it is optimal to specify a stake for the borrower large enough that the incentive constraint (IC_b) is satisfied. Because condition (3.8) implies that the project has negative NPV in the case of misbehavior, such a specification is clearly optimal when the contract cannot be renegotiated. The purpose of this exercise is to check in a rather mechanical way that the borrower cannot gain by offering a loan agreement in which (IC_b) is not satisfied, and which is potentially renegotiated before the borrower chooses her effort. While there is a more direct way to prove this result, some insights are gleaned from this pedestrian approach. Indeed, the exercise provides conditions under which the lender is willing to forgive debt in order to boost incentives (the analysis will bear some resemblance to that of liquidity shocks in Chapter 5, except that the lender's concession takes the form of debt forgiveness rather than cash infusion).[60]

(i) Consider a loan agreement specifying investment I and stake $R_b < BI/\Delta p$ for the borrower. Suppose that the loan agreement can be renegotiated after it is signed and the investment is sunk and before the borrower chooses her effort. Renegotiation takes place if and only if it is mutually advantageous. Show that the loan agreement is renegotiated if and only if

$$(\Delta p)RI - \frac{p_H BI}{\Delta p} + p_L R_b \geqslant 0.$$

(ii) Interpret the previous condition. In particular, show that it can be obtained directly from the general theory. Hint: consider a fictitious, "fixed-investment" project with income $(\Delta p)RI$, investment 0, and cash on hand $p_L R_b$.

(iii) Assume for instance that the entrepreneur makes a take-it-or-leave-it offer in the renegotiation (that is, the entrepreneur has the bargaining power). Compute the borrowing capacity when $R_b < BI/\Delta p$ and the loan agreement is renegotiated.

(iv) Use a direct, rational expectations argument to point out in a different way that there is no loss of generality in assuming $R_b \geqslant BI/\Delta p$ (and therefore no renegotiation).

Exercise 3.7 (strategic leverage). (i) A borrower has assets A and must find financing for an investment $I(\tau) > A$. As usual, the project yields R (success) or 0 (failure). The borrower is protected by limited liability. The probability of success is $p_H + \tau$ or $p_L + \tau$, depending on whether the borrower works or shirks, with $\Delta p = p_H - p_L > 0$. There is no private benefit when working and private benefit B when shirking. The financial market is competitive and the expected rate of return demanded by investors is equal to 0. It is never optimal to give incentives to shirk.

60. The phenomenon of debt renegotiation has been analyzed in a number of settings: see, for example, Bulow and Rogoff (1989a,b), Eaton and Gersovitz (1981), Fernandez and Rosenthal (1990), Gale and Hellwig (1989), Gromb (1994), Hart and Moore (1989, 1995), and Snyder (1994).

The investment cost I is an increasing and convex function of τ (it will be further assumed that $p_H R > I(0)$, that in the relevant range $p_H + \tau < 1$, and that $I'(0)$ is "small enough" so as to guarantee an interior solution). Let τ^*, A^*, and τ^{**} be defined by

$$I'(\tau^*) = R,$$
$$[p_H + \tau^*]\left[R - \frac{B}{\Delta p}\right] = I(\tau^*) - A^*,$$
$$I'(\tau^{**}) = R - \frac{B}{\Delta p}.$$

Can the borrower raise funds? If so, what is the equilibrium level τ of "quality of investment"?

(ii) Suppose now that there are two firms (that is, two borrowers) competing on this product market. If only firm i succeeds in its project, its income is (as in question (i)), equal to R (and firm j's income is 0). If the two firms succeed (both get hold of "the technology"), they compete *à la* Bertrand in the product market and get 0 each. For simplicity, assume that the lenders observe only whether the borrower's income is R or 0, rather than whether the borrower has succeeded in developing the technology (showoffs: you can discuss what would happen if the lenders observed "success/failure"!).

So, if $q_i \equiv p_i + \tau_i$ denotes the probability that firm i develops the technology (with $p_i = p_H$ or p_L), the probability that firm i makes R is $q_i(1 - q_j)$. (This assumes implicitly that projects are independent.)

Consider the following timing. (1) Each borrower simultaneously and secretly arranges financing (if feasible). A borrower's leverage (or quality of investment) is *not* observed by the other borrower. (2) Borrowers choose whether to work or shirk. (3) Projects succeed or fail.

- Let $\hat{\tau}$ be defined by

$$I'(\hat{\tau}) = [1 - (p_H + \hat{\tau})]R.$$

  Interpret $\hat{\tau}$.
- Suppose that the two borrowers have the same initial net worth A. Find the lower bound $\hat{A}$ on A such that $(\hat{\tau}, \hat{\tau})$ is the (symmetric) Nash outcome.
- Derive a sufficient condition on A under which it is an equilibrium for a single firm to raise funds.

(iii) Consider the set up of question (ii), except that borrower 1 moves first and publicly chooses τ_1. Borrower 2 may then try to raise funds (one will assume either that τ_2 is secret or that borrower 1 is rewarded on the basis of her success/failure performance; this is in order to avoid strategic choices by borrower 2 that would try to induce borrower 1 to shirk). Suppose that each has net worth $\tilde{A}$ given by

$$\tilde{q}\left[(1 - \tilde{q})R - \frac{B}{\Delta p}\right] = I(\tilde{q} - p_H) - \tilde{A},$$

where $\tilde{q}$ satisfies

$$I'(\tilde{q} - p_H) = (1 - \tilde{q})R - \frac{B}{\Delta p}.$$

- Interpret $\tilde{q}$.
- Show that it is optimal for borrower 1 to choose $\tau_1 > \tilde{q} - p_H$.

Exercise 3.8 (equity multiplier and active monitoring). (i) Derive the equity multiplier in the variable-investment model. (Reminder: the investment $I \in [0, \infty)$ yields income RI in the case of success and 0 in the case of failure. The borrower's private benefit from misbehaving is equal to BI. Misbehaving reduces the probability of success from p_H to $p_L = p_H - \Delta p$. The borrower has cash A and is protected by limited liability. Assume that $\rho_1 = p_H R > 1$, $\rho_0 = p_H(R - B/\Delta p) < 1$ and $1 > p_L R + B$. The investors' rate of time preference is equal to 0.) Show that the equity multiplier is equal to $1/(1 - \rho_0)$.

(ii) Derive the equity multiplier with active monitoring: the entrepreneur can hire a monitor, who, at private cost cI, reduces the entrepreneur's private benefit from shirking from BI to $b(c)I$, where $b(0) = B$, $b' < 0$. The monitor must be given incentives to monitor (denote by R_m his income in the case of success). The monitor wants to break even, taking into account his private monitoring cost (so, there is "no shortage of monitoring capital").

- Suppose that the entrepreneur wants to induce level of monitoring c. Write the two incentive constraints to be satisfied by R_m and R_b (where R_b is the borrower's reward in the case of success).
- What is the equity multiplier?
- Show that the entrepreneur chooses c so as to maximize

$$\max_c \left\{ \frac{\rho_1 - 1 - c}{1 - \rho_0 + (p_H/\Delta p)[b(c) + c - B]} \right\}.$$

Exercise 3.9 (concave private benefit). Consider the variable-investment model with a concave private benefit. The entrepreneur obtains $B(I)$ when shirking and 0 when behaving, where $B(0) = 0$, $B' > 0$, $B'' < 0$ (and $B'(0)$ large, $\lim_{I\to\infty} B'(I) = B$, where $p_H(R - B/\Delta p) < 1$).

(i) Compute the borrowing capacity.

(ii) How does the shadow price v of the entrepreneur's cash on hand vary with A?

Exercise 3.10 (congruence, pledgeable income, and power of incentive scheme). The credit rationing model developed in this chapter assumes that the entrepreneur's and investors' interests are *a priori* dissonant, and that incentives must be aligned by giving the entrepreneur enough of a stake in the case of success.

Suppose that the entrepreneur and the investors have indeed dissonant preferences with probability x, but have naturally aligned interests with probability $1 - x$. Which prevails is unknown to both sides at the financing stage and is discovered (only) by the entrepreneur just before the moral-hazard stage.

More precisely, consider the fixed-investment model of Section 3.2. The investors's outlay is $I - A$ and they demand an expected rate of return equal to 0. The entrepreneur is risk neutral and protected by limited liability. With probability x, interests are dissonant: the entrepreneur obtains private benefit B by misbehaving (the probability of success is p_L) and 0 by behaving (probability of success p_H). With probability $1 - x$, interests are aligned: the entrepreneur's taking her private benefit B coincides with choosing probability of success p_H.

(i) Consider a "simple incentive scheme" in which the entrepreneur receives R_b in the case of success and 0 in the case of failure. R_b thus measures the "power of the incentive scheme."

Show that it may be optimal to choose a low-powered incentive scheme if preferences are rather congruent (x low) and that the incentive scheme is necessarily high-powered if preferences are rather dissonant (x high).

(ii) Show that one cannot improve on simple incentive schemes by presenting the entrepreneur with a menu of two options (two outcome-contingent incentive schemes) from which she will choose once she learns whether preferences are congruent or dissonant.

Exercise 3.11 (retained-earnings benefit). An entrepreneur has at date 1 a project of fixed size with characteristics $\{I^1, R^1, p_H^1, p_L^1, B^1\}$ (see Section 3.2). This entrepreneur will at date 2 have a different fixed size project with characteristics $\{I^2, R^2, p_H^2, p_L^2, B^2\}$, which will then require new financing. So, we are considering only a short-term loan for the first project. Retained earnings from the first project can, however, be used to defray part of the investment cost of the second project. Assume that all the characteristics of the second project are known at date 1 except B^2, which is distributed on $[\underline{B}^2, \bar{B}^2]$ according to the cumulative distribution $F(B^2)$. Assume for simplicity that $\underline{B}^2 > \Delta p^2(p_H^2 R^2 - I^2)/p_H^2$. The characteristics of the second project become common knowledge at the beginning of date 2.

(i) Compute the shadow value of retained earnings. (Hint: what is the entrepreneur's gross utility in period 2?)

(ii) Show that it is possible that the first project is funded even though it would not be funded if the second project did not exist and even though the entrepreneur cannot pledge at date 1 income resulting from the second project.

Exercise 3.12 (investor risk aversion and risk premia). One of the key developments in the theory of *market* finance has been to find methods to price claims held by investors. Market finance emphasizes state-contingent pricing, the fact that 1 unit of income does not have a uniform value across states of nature. This book assumes that investors are risk neutral, and so it does not matter how the pledgeable income is spread across states of nature. This assumption is made only for the sake of computational simplicity, and can easily be relaxed.

Consider a two-date model of market finance with a representative consumer/investor. This consumer has utility of consumption $u(c_0)$ at date 0, the date at which he lends to the firm, and utility of consumption $u(c(\omega))$ at date 1, date at which he receives the return from investment. There is macroeconomic uncertainty in that the representative consumer's date-1 consumption depends on the state of nature ω. The state of nature describes both what happens

in this particular firm and in the rest of the economy (even though aggregate consumption is independent of the outcome in this particular firm to the extent that the firm is atomistic, which we will assume).

Suppose that the entrepreneur works. Let S denote the event "the project succeeds" and F the event "the project fails." Let

$$q_S = E\left[\frac{u'(c(\omega))}{u'(c_0)} \;\middle|\; \omega \in S\right]$$

and

$$q_F = E\left[\frac{u'(c(\omega))}{u'(c_0)} \;\middle|\; \omega \in F\right].$$

The firm's activity is said to covary positively with the economy (be "procyclical") if $q_S < q_F$, and negatively (be "countercyclical") if $q_F < q_S$.

Suppose that

$$p_H q_S + (1 - p_H) q_F = 1.$$

(i) Interpret this assumption.

(ii) In the fixed-investment model of Section 3.2 (and still assuming that the entrepreneur is risk neutral), derive the necessary and sufficient condition for the project to receive financing.

(iii) What is the optimal contract between the investors and the entrepreneur? Does it involve maximum punishment ($R_b = 0$) in the case of failure? How would your answer change if the entrepreneur were risk averse? (For simplicity, assume that her only claim is in the firm. She does not hold any of the market portfolio.)

Exercise 3.13 (lender market power). (i) *Fixed investment.* An entrepreneur has cash amount A and wants to invest $I > A$ into a (fixed-size) project. The project yields $R > 0$ with probability p and 0 with probability $1 - p$. The probability of success is p_H if the entrepreneur works and $p_L = p_H - \Delta p$ ($\Delta p > 0$) if she shirks. The entrepreneur obtains private benefit B if she shirks and 0 otherwise. The borrower is protected by limited liability and everyone is risk neutral. The project is worthwhile only if the entrepreneur behaves.

There is a single lender. This lender has access to funds that command an expected rate of return equal to 0 (so the lender would content himself with a 0 rate of return, but he will use his market power to obtain a superior rate of return). Assume

$$V \equiv p_H R - I > 0$$

and let $\bar{A}$ and $\hat{A}$ be defined by

$$p_H\left[R - \frac{B}{\Delta p}\right] = I - \bar{A}$$

and

$$p_H \frac{B}{\Delta p} - \hat{A} = 0.$$

Assume that $\bar{A} > 0$ and that the *lender* makes a take-it-or-leave-it offer to the borrower (i.e., the lender chooses R_b, the borrower's compensation in the case of success).

- What contract is optimal for the lender?
- Is the financing decision affected by lender market power (i.e., compared with the case of competitive lenders solved in Section 3.2)?
- Draw the borrower's *net* utility (i.e., net of A) as a function of A and note that it is nonmonotonic (distinguish four regions: $(-\infty, \bar{A})$, $[\bar{A}, \hat{A})$, $[\hat{A}, I)$, $[I, \infty)$). Explain.

(ii) *Variable investment.* Answer the first two bullets in question (i) (lender's optimal contract and impact of lender market power on the investment decision) in the variable-investment version. In particular, show that lender market power reduces the scale of investment. (Reminder: I is chosen in $[0, \infty)$. The project yields RI if successful and 0 if it fails. Shirking, which reduces the probability of success from p_H to p_L, yields private benefit BI. Assume that $p_H R > 1 > p_H(R - B/\Delta p)$. Hint: show that the two constraints in the lender's program are binding.)

Exercise 3.14 (liquidation incentives). This exercise extends the fixed-investment model of Section 3.2 by adding a signal on the profitability of the project that (a) accrues after effort has been chosen, and (b) is privately observed. (The following model is used as a building block in a broader context by Dessi (2005).)

An entrepreneur has cash A and wants to invest $I > A$ into a project. The project yields R (success) or 0 (failure) at the end. An intermediate signal reveals the probability γ that the project will succeed, with $\gamma = \bar{\gamma}$ or $\underline{\gamma}$ ($\bar{\gamma} = \underline{\gamma} + \Delta\gamma$ and $\Delta\gamma > 0$). The probability, p, that $\gamma = \bar{\gamma}$ depends on the entrepreneur's effort. If the entrepreneur behaves, then $p = p_H$ and the entrepreneur receives no private benefit. If the entrepreneur misbehaves, then $p = p_L$

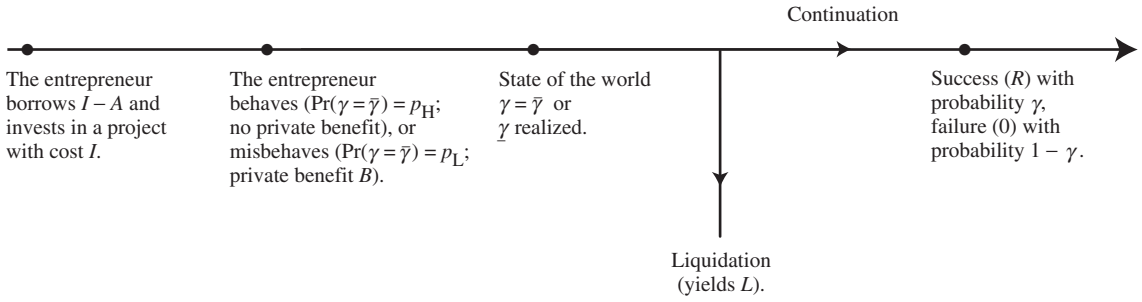


Figure 3.11

and the entrepreneur receives private benefit B. Investors and entrepreneur are risk neutral and the latter is protected by limited liability. The competitive rate of return is equal to 0.

Introduce further an option to liquidate after the signal is realized but before the final profit accrues. Liquidation yields L, and L is entirely pledgeable to investors.

One will assume that

$$\bar{\gamma}R > L > \underline{\gamma}R,$$

so that it is efficient to liquidate if and only if the signal is bad; and that

$$p_H\bar{\gamma}R + (1 - p_H)L > I$$

(which will imply that the NPV is positive).

Figure 3.11 summarizes the timing.

(i) Suppose first that γ is verifiable. Argue that the entrepreneur should be rewarded solely as a function of the realization of γ. What is the pledgeable income? Show that the project is financed if and only if $A \geqslant \bar{A}$, where

$$p_H\left(\bar{\gamma}R - \frac{B}{\Delta p}\right) + (1 - p_H)L = I - \bar{A}.$$

(ii) Suppose now that γ is observed only by the entrepreneur. This implies that the entrepreneur must be induced to tell the truth about γ. Without loss of generality, consider an incentive scheme in which the entrepreneur receives R_b in the case she announces $\gamma = \bar{\gamma}$ (and therefore the project continues) *and* the final profit is R, L_b if she announces $\gamma = \underline{\gamma}$ (and therefore the project is liquidated), and 0 otherwise.

Show that the project is funded if and only if

$$A \geqslant \bar{A} + \underline{\gamma}\frac{B}{(\Delta p)(\Delta\gamma)}.$$

Exercise 3.15 (project riskiness and credit rationing). Consider the basic, fixed-investment model (the investment is I, the entrepreneur borrows $I - A$; the probability of success is p_H (no private benefit) or $p_L = p_H - \Delta p$ (private benefit B), success (failure) yields verifiable profit R (respectively 0)). There are two variants, "A" and "B," of the projects, which differ only with respect to "riskiness":

$$p_H^A R^A = p_H^B R^B, \quad \text{but } p_H^A > p_H^B;$$

so project B is "riskier." The investment cost is the same for both variants and, furthermore,

$$p_H^A - p_L^A = p_H^B - p_L^B.$$

Which variant is less prone to credit rationing?

Exercise 3.16 (scale versus riskiness tradeoff). Consider an entrepreneur with a project of variable investment I. The entrepreneur has initial wealth A, is risk neutral, and is protected by limited liability. Investors are risk neutral and demand a rate of return equal to 0.

The project comes in two versions:

Risky. The project costs I and ends up (potentially) productive only with probability $x < 1$. The timing goes as follows. (a) The scale of investment I is selected. (b) After the investment has been sunk, news accrues as to the profitability of the project. With probability $1 - x$, the project stops and yields 0. With probability x, the project continues (without any need for reinvestment). In the latter case, (c) the entrepreneur chooses an effort; good behavior

confers no private benefit on the entrepreneur and yields subsequent probability of success p_H; misbehavior confers private benefit BI and yields probability of success p_L. Finally, (d) the outcome accrues: success yields RI and failure 0.

Safe. The investment cost, XI with $X > 1$, is higher for a given size I. But the project is always productive ("$\varkappa = 1$"). The moral hazard and outcome stages are as in the case of a risky choice.

We will assume that the contract aims at inducing good behavior. Letting

$$\rho_1 \equiv p_H R \quad \text{and} \quad \rho_0 \equiv p_H\left(R - \frac{B}{\Delta p}\right),$$

one will further assume that $\varkappa > 1/\rho_1$ and $X < \rho_1$.

Assume that entrepreneur and investors contract on which version will be selected.

(i) Show that the risky version is chosen if and only if

$$\varkappa X \geqslant 1.$$

(ii) Interpret this condition in terms of a "cost of bringing 1 unit of investment to completion."

Exercise 3.17 (competitive product market interactions). There is a mass 1 of identical entrepreneurs with the variable-investment technology described in Section 3.4. The representative entrepreneur has wealth A, is risk neutral, and is protected by limited liability.

Denote the average investment by I and the individual investment i (in equilibrium $i = I$ by symmetry but we need to distinguish the two in a first step in order to compute the competitive equilibrium). A project produces Ri units of goods when successful and 0 when it fails. The probability of success is p_H in the case of good behavior (the entrepreneur receives no private benefit) and $p_L = p_H - \Delta p$ in the case of misbehavior (the entrepreneur then receives private benefit Bi). Assume that it is optimal to induce the entrepreneur to behave.

The market price of output is $P = P(Q)$, with $P' < 0$, where Q is aggregate production (with $P(Q)$ tending to 0 as Q goes to infinity, to ensure that aggregate investment is finite). Finally, the shocks faced by the firms are independent (there is no industry-wide uncertainty) and the risk-neutral investors demand a rate of return equal to 0.

Show that the equilibrium is unique. Compute the equilibrium level of investment. (Hint: distinguish two cases, depending on whether A is large or small.)

Exercise 3.18 (maximal incentives principle in the fixed-investment model). Pursue the analysis of Section 3.4.3, but for the fixed-investment model of Section 3.2: the investment cost I is given and the income is either R^S or R^F (instead of R or 0), where $R^S > R^F > 0$. We assume that

$$R^F < I - A,$$

so the project cannot be straightforwardly financed by bringing in net worth A and pledging the lower income R^F to lenders. Let

$$R \equiv R^S - R^F$$

denote the increase in income from the low to the high level. Show that the debt contract is optimal, but unlike in the variable-investment case it may not be uniquely optimal.

Exercise 3.19 (balanced-budget investment subsidy and profit tax). This exercise shows that a balanced-budget public policy that is not based on information that is superior to investors' does not boost pledgeable income and therefore outside financing capacity (unless there are externalities among firms: see Exercise 3.17). This general point is illustrated in the context of the variable-investment model: an entrepreneur has cash amount A and wants to invest $I > A$ into a (variable size) project. The project yields $RI > 0$ with probability p and 0 with probability $1 - p$. The probability of success is p_H if the entrepreneur works and $p_L = p_H - \Delta p$ ($\Delta p > 0$) if she shirks. The entrepreneur obtains private benefit BI if she shirks and 0 otherwise. The borrower is protected by limited liability and everyone is risk neutral. The project is worthwhile only if the entrepreneur behaves. Competitive lenders demand a zero expected rate of return. Assume that the NPV is positive:

$$\rho_1 \equiv p_H R > 1,$$

but

$$\rho_0 \equiv p_H\left(R - \frac{B}{\Delta p}\right) < 1.$$

The government has two instruments: a subsidy s per unit of investment, and a proportional tax t on the final profit.

The government must set (s,t) so as to balance its budget. Show that the government's policy is neutral:

$$I = \frac{A}{1-\rho_0} \quad \text{and} \quad U_b = (\rho_1 - 1)I$$

for any (s,t), where U_b is the entrepreneur's utility.

Exercise 3.20 (variable effort, the marginal value of net worth, and the pooling of equity). In the fixed-investment model, the shadow price of entrepreneurial net worth is equal to 0 almost everywhere and is infinite at the threshold $A = \bar{A}$. A more continuous response arises when the entrepreneur's effort is continuous rather than discrete. The object of this exercise is to show that the shadow price is positive and *decreasing in A* in the range in which the entrepreneur is able to finance her project but must borrow from investors. It then applies the analysis to the internal allocation of funds between two divisions.

An entrepreneur has cash A and wants to invest $I > A$ into a fixed-size project. The project yields R with probability p and 0 with probability $1-p$. Reaching a probability of success p requires the entrepreneur to sink (unobservable) effort cost $\frac{1}{2}p^2$ (there is no private benefit in this version). The borrower is risk neutral and is protected by limited liability. Investors are risk neutral and the market rate of interest is 0. Assume that $\sqrt{2I} < R < 1$.

(i) Note that, had the borrower no need to borrow ($A \geqslant I$), the borrower's net utility would be

$$U_b = V^* = \tfrac{1}{2}R^2 - I,$$

independently of A.

(ii) Find the threshold $\bar{A}$ under which the project is not funded. (Hint: write the pledgeable income as a function of the entrepreneur's reward R_b in the case of success. Argue that one can focus attention on the values of R_b that exceed $\frac{1}{2}R$. Do not forget that the NPV must be nonnegative.)

Letting $V(A)$ denote the NPV in the region in which the entrepreneur's project is financed. Show that the shadow price of net worth, $V'(A)$, satisfies

$$V'(A) > 0,$$
$$V'(I) = 0,$$
$$V''(A) < 0.$$

(iii) Following Cestone and Fumagalli (2005), consider two entrepreneurs, each with net worth A. They will each have a project described as above, but with random investment cost. For simplicity, one of them will face investment cost I_H and the other I_L, where

$$I_L - A < \tfrac{1}{4}R^2 < I_H - A,$$

but it is not known in advance who will face which investment cost (each is equally likely to be the lucky entrepreneur). Investment costs, however, will become publicly known before the investments are sunk. Assume that

$$\tfrac{3}{8}R^2 > I_H,$$

so that the only binding constraint for financing in question (ii) is the investors' breakeven constraint; and that

$$\tfrac{1}{2}R^2 > (I_L + I_H) - 2A,$$

and so both projects can be financed by pooling resources. Do the entrepreneurs, behind the veil of ignorance, want to pool their resources and commit to force the lucky firm to cross-subsidize the unlucky one? (Hint: show that under pooling, and, if both invest, the net worth is split so that both entrepreneurs have the same stake in success.)

Exercise 3.21 (hedging or gambling on net worth?). Froot et al. (1993) analyze an entrepreneur's risk preferences with respect to net worth. In the notation of this book, the situation they consider is summarized in Figure 3.12.

The entrepreneur is risk neutral and protected by limited liability. The investors are risk neutral and demand a rate of return equal to 0.

At date 0, the entrepreneur decides whether to insure against a date-1 income risk

$$r = A_0 + \varepsilon,$$

where $\varepsilon \in [\underline{\varepsilon}, \bar{\varepsilon}]$, $E[\varepsilon] = 0$, and $A_0 + \underline{\varepsilon} \geqslant 0$.

For simplicity, we allow only a choice between full hedging and no hedging (the theory extends straightforwardly to arbitrary degrees of hedging). Hedging (which wipes out the noise and thereby guarantees that the entrepreneur has cash on hand A_0 at date 1) is costless.

After receiving income, the entrepreneur uses her cash to finance investment I and must borrow $I - A$ from investors, with $A = A_0$ in the case of hedging and $A = A_0 + \varepsilon$ in the absence of hedging (provided that $A \leqslant I$; otherwise there is no need to borrow).

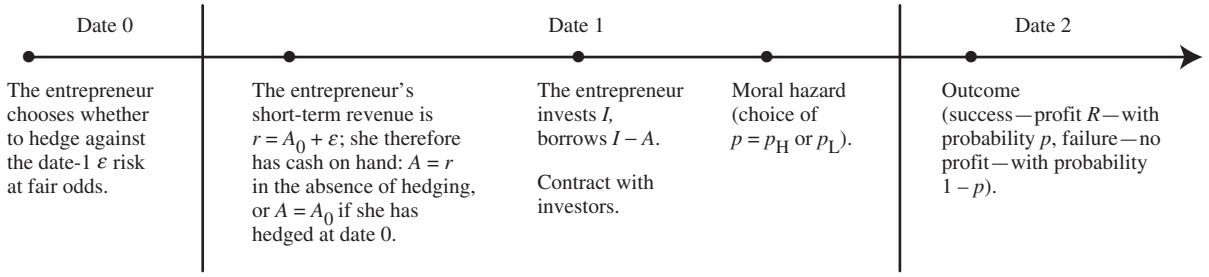


Figure 3.12

Note that there is no overall liquidity management as there is no contract at date 0 with the financiers as to the future investment.

This exercise investigates a variety of situations under which the entrepreneur may prefer either hedging or "gambling" (here defined as "no hedging").

(i) *Fixed investment, binary effort.* Suppose that the investment size is fixed (as in Section 3.2), and that the entrepreneur at date 1, provided that she receives funding, either behaves (probability of success p_H, no private benefit) or misbehaves (probability of success p_L, private benefit B). As usual, the project is not viable if it induces misbehavior and has a positive NPV ($p_H R > I > p_L R + B$, where R is the profit in the case of success). Let $\bar{A}$ be defined (as in Section 3.2) by

$$p_H\left(R - \frac{B}{\Delta p}\right) = I - \bar{A}.$$

Suppose that ε has a wide support.

Show that the entrepreneur

- hedges if $A_0 \geqslant \bar{A}$,
- gambles if $A_0 < \bar{A}$.

(ii) *Fixed investment, continuous effort.* Suppose, as in Exercise 3.20, that succeeding with probability p involves an unverifiable private cost $\frac{1}{2}p^2$ for the entrepreneur (so, effort in this subquestion involves a cost rather than the loss of a private benefit). (Assume $R < 1$ to ensure that probabilities are smaller than 1.)

Write the investors' breakeven condition as well as the NPV as functions of the entrepreneur's stake, R_b, in success. Note that one can focus without loss of generality on $R_b \in [\frac{1}{2}R, R]$. Assume that $I - A_0 < \frac{1}{4}R^2$ and that the support of ε is sufficiently small that the entrepreneur always receives funding when she does not hedge (and *a fortiori* when she hedges). This assumption eliminates the concerns about financing of investment that were crucial in question (i).

Show that the entrepreneur hedges.

(iii) *Variable investment.* Return to the binary effort case ($p = p_H$ or p_L), but assume that the investment I is variable (as in Section 3.4). The income is RI in the case of success and 0 in the case of failure. The private benefit of misbehaving is $B(I)$ with $B' > 0$. Assume that the size of investment is always constrained by the pledgeable income and that the optimal contract induces good behavior.

Show that the entrepreneur

- hedges if $B(\cdot)$ is convex;
- is indifferent between hedging and gambling if $B(\cdot)$ is linear;
- gambles if $B(\cdot)$ is concave.

(iv) *Variable investment and unobservable income.* Suppose that the investment size is variable and that the income from investment $R(I)$ is unobservable by investors (fully appropriated by the entrepreneur) and is concave. Suppose that it is always optimal for the entrepreneur to invest her cash on hand.

Show that the entrepreneur hedges.

(v) *Liquidity and risk management.* Suppose, in contrast with Froot et al.'s analysis, that the entrepreneur can sign a contract with investors at date 0. Show that the entrepreneur's utility can be maximized by insulating the date-1 volume of investment from the realization of ε, i.e., with full hedging, even in situations where gambling was optimal when funding was secured only at date 1.

Lastly, the supplementary section investigates the rationales for group lending, which turn out to be closely related to some of the themes of this chapter. The supplementary section argues that group lending may be an attempt either to use social capital as collateral or to use peer monitoring in order to reduce agency costs.

4.2 Boosting the Ability to Borrow: Diversification and Its Limits

The computation of the equity multiplier in Section 3.4 was conducted under the assumption that the probability of success is independent of the scale of the investment. As we will observe, this implicitly assumed that if an expansion in the scale of "the project" actually stands for an increase in the number of projects, then the projects' outcomes are perfectly correlated (conditional on the effort that is exerted on them).[2] This formalization depicted a polar case in which there are no benefits to diversification.

However, as Diamond (1984) has forcefully argued,[3] diversification may bring substantial incentive benefits when projects are independent. Intuitively, the borrower can *cross-pledge* the incomes of various projects. That is, she can use the income she receives on a successful project as *collateral* for other projects. Such cross-pledging is useless when projects are correlated, because, when a project fails, the collateral posted for this project (the income from other projects) is valueless.

We analyze the incentive benefits from diversification in the cases of two independent projects and of a large number of such projects.[4] We then point at some limits to the diversification argument.

4.2.1 The Benefits of Diversification: The Case of Two Projects

Let us consider two independent and identical projects with fixed investment size I. That is, the two projects are as described in Section 3.2. Projects succeed (yield R) or fail (yield 0). The probability of success is p_H if the entrepreneur behaves (but then receives no private benefit) and p_L if she misbehaves (and receives private benefit B). Let $2A$ denote the entrepreneur's initial wealth, that is, A per project. The borrower is risk neutral and protected by limited liability. The lenders are risk neutral and demand an expected rate of return equal to 0.

If both projects are funded, then the borrower can work on both, shirk on both, or work on either of them. There can also be four outcomes: both projects succeed, they both fail, or only one of them succeeds. It is clear that two projects are undertaken only if the incentive scheme induces the borrower to work on both projects. Otherwise, the borrower would be better off undertaking one project or none.

4.2.1.1 Project Financing

Let us begin with the benchmark of stand-alone financing for each project. Project financing refers to the provision of funding for a given, well-identified project. The analysis is then that of Section 3.2 for

2. One way to think about the case of perfect correlation is to introduce a latent random variable ω that is, say, uniformly distributed on $[0,1]$ and that is realized after the borrower's choice of efforts on the various projects. If $0 \leqslant \omega < p_L$, then a project succeeds even if the borrower shirked on the project. If $\omega \geqslant p_H$, a project fails even if the borrower worked on the project. Lastly, if $p_L \leqslant \omega < p_H$, a project succeeds if and only if the borrower worked on the project. Note that the latent variable is the same for all projects.

The model with multiple projects with sizes $I_1, \ldots, I_n$ and private benefits $B_1, \ldots, B_n$ can then be shown to be equivalent to a model with a single project with size $I = \sum_i I_i$ and private benefit $B = \sum_i B_i$. Heuristically, the pledgeable income is the same in the case of multiple projects and of a single, large project.

The reader might intuit that the borrower has more leeway for misbehavior in the case of multiple projects, as she has other alternatives (shirk on some projects and work on others) to working on all projects than shirking on all projects. This intuition, however, is misleading because these "partial deviations" are perfectly detected whenever they might be beneficial. Indeed, suppose $\omega < p_L$ (respectively, $\omega \geqslant p_H$). Then all projects succeed (respectively, fail) regardless of effort and the borrower would be better off shirking on all projects. And if $p_L \leqslant \omega < p_H$, some projects succeed and some fail, proving unambiguously that the borrower has deviated from the strategy of zealousness on all projects. Thus, if the borrower receives nothing in such situations, the best strategy for the borrower is, as in the case of a single project, either to work (on all projects) or to shirk (on all projects). Even with multiple projects, there is a single relevant incentive constraint (work or shirk).

In contrast, the case of independent projects can be represented by a set of independent random variables $\{\omega_i\}_{i=1,\ldots,n}$ with the same distribution as ω.

3. See, for example, Cerasi and Daltung (2000), Matutes and Vives (1996), Williamson (1986), and Yanelle (1989) for contributions that make use of Diamond's argument.

4. Similar expositions of the Diamond argument can be found in Holmström (1993) and Hellwig (2000).

each project taken in isolation. The borrower receives R_b in the case of success and 0 in the case of failure of a given project, independently of what happens in the borrower's other activity. As usual, the incentive constraint for a given project is

$$(\Delta p)R_b \geqslant B,$$

and the per-project financing condition is that the pledgeable income exceeds the investors' initial outlay:

$$p_H\left(R - \frac{B}{\Delta p}\right) \geqslant I - A$$

or

$$A \geqslant \bar{A}.$$

This condition can be interpreted as a capital or net worth requirement. If $A < \bar{A}$, project financing is not viable.

Note that project financing does not make full use of the borrower's potential liability. When project 1, say, fails, then with conditional probability p_H (which is the prior probability under statistical independence of the two projects), project 2 is successful and returns R_b to the entrepreneur. Even under limited liability the entrepreneur's income on the first project can be brought down to $[-R_b]$ (conditional on the second project succeeding) rather than 0. We now make use of this observation.

4.2.1.2 Cross-Pledging

Let us now bring the two projects under a single roof (a "firm"), or at least allow joint liability between the two projects, so that the income on one project is used as collateral for the other project. Let R_2, R_1, R_0 denote the borrower's reward when the number of successful projects is 2, 1, 0, respectively. A risk-neutral borrower cares only about her expected reward, and thus the loan agreement should be structured so as to provide the borrower with maximal incentives for a given expected reward

$$p_H^2 R_2 + 2p_H(1-p_H)R_1 + (1-p_H)^2 R_0.$$

Intuitively, this requires that the borrower be rewarded only when the two projects are successful, namely, $R_2 > 0$, $R_1 = R_0 = 0$ (or, more precisely, there always exists *one* optimal incentive scheme which rewards the borrower only in the case of full success). Showing this formally is a simple exercise, which we leave to the reader,[5] who can also consult Section 4.7 for a closely related result. Note that $R_1 = 0$ corresponds to full cross-pledging (contrast this with project financing, under which $R_1 = R_b > 0$, where R_b is the entrepreneur's compensation in the case of success in a given project).

Taking this feature of the incentive scheme for granted, the condition that guarantees that the borrower prefers to work on both projects to working on neither is

$$p_H^2 R_2 - 2B \geqslant p_L^2 R_2$$

or

$$(p_H + p_L)R_2 \geqslant 2\frac{B}{\Delta p}. \tag{4.1}$$

Note that this condition implies that the borrower also prefers to work on both projects to working on a single one: by shirking on the second project, say, the borrower reduces the probability of full success by p_H (the probability that the first project succeeds) times Δp (the reduction in the second project's probability of success). And thus the second incentive constraint can be written as

$$p_H(\Delta p)R_2 \geqslant B. \tag{4.2}$$

Since $p_H > \frac{1}{2}(p_H + p_L)$, this second constraint (4.2) is automatically satisfied if the first, (4.1), is.

Let us now compute the expected pledgeable income. It is equal to the expected return on the projects, $2p_H R$, minus the minimum expected payoff to the borrower, $p_H^2 R_2$, that is consistent with incentive compatibility. From (4.1) the latter is

$$p_H^2 R_2 = \frac{2p_H^2 B}{(p_H + p_L)\Delta p} = 2(1-d_2)\frac{p_H B}{\Delta p},$$

---

5. There are two incentive constraints. First, the borrower must prefer to work on both projects to working on a single one, and so

$$p_H^2 R_2 + 2p_H(1-p_H)R_1 + (1-p_H)^2 R_0 - 2B \geqslant p_H p_L R_2 + (p_H + p_L - 2p_H p_L)R_1 + (1-p_H)(1-p_L)R_0 - B.$$

She must also prefer working on both projects to working on none, and so

$$p_H^2 R_2 + 2p_H(1-p_H)R_1 + (1-p_H)^2 R_0 - 2B \geqslant p_L^2 R_2 + 2p_L(1-p_L)R_1 + (1-p_L)^2 R_0.$$

It then suffices to show that for a given $\{R_2, R_1, R_0\}$ satisfying these two inequalities, there exists R_2' such that $\{R_2', 0, 0\}$ also satisfies the two inequalities and provides the entrepreneur with the same expected compensation:

$$p_H^2 R_2' = p_H^2 R_2 + 2p_H(1-p_H)R_1 + (1-p_H)^2 R_0.$$

where

$$d_2 \equiv \frac{p_L}{p_L + p_H} \in (0, \tfrac{1}{2})$$

is an agency-based measure of economies of diversification into two independent projects. Letting $2A$ denote the borrower's initial net worth (so, A is her per-project cash on hand), the two projects can be funded if

$$2p_H R - 2(1 - d_2)\frac{p_H B}{\Delta p} \geqslant 2I - 2A,$$

or

$$p_H\left[R - (1 - d_2)\frac{B}{\Delta p}\right] \geqslant I - A, \tag{4.3}$$

or

$$A \geqslant \bar{\bar{A}}, \quad \text{with } \bar{\bar{A}} \equiv I - p_H\left[R - (1 - d_2)\frac{B}{\Delta p}\right] < \bar{A}.$$

Thus, cross-pledging facilitates financing.

Role of correlation. The benefits from cross-pledging come from the diversification effect. We have assumed that projects were independent. Suppose, in contrast, that the two projects are perfectly correlated. Then, condition (3.3) implies that they can both be funded if and only if

$$p_H\left[R - \frac{B}{\Delta p}\right] \geqslant I - A \quad \text{or} \quad A \geqslant \bar{A}.$$

In words, there is no cost to project financing if projects are perfectly correlated. Or, put differently, the effect of diversification, that is, of the independence of the two projects, is tantamount to a reduction of the private benefit from B to $(1 - d_2)B$. Because of the independence of the two projects, the borrower can pledge his income on a project as collateral for the other project, were the second project to fail. Thus *project finance*, namely, a mode of financing that establishes (unrelated) claims on individual projects, *is here suboptimal* unless $d_2 = 0$, that is, unless there are no economies of diversification. We refer to Exercise 4.4 for the study of arbitrary (positive or negative) correlation between the two projects.

Variable investment size. In the case of fixed investment sizes, the benefit from diversification takes the form of a facilitated access to financing. Conditional on getting financing, the total NPV ($2(p_H R - I)$) is, of course, unchanged. With variable investment sizes, the extent of financing, rather than the access to financing, is the issue. Then diversification increases the borrowing capacity and therefore the NPV (see Exercise 4.10).

4.2.2 The Benefits of Diversification: A Large Number of Projects

The previous diversification result extends straightforwardly to n independent projects.

For the purpose of this section, let us assume that

$$p_H R - I < B.$$

The reader will check that a borrower with net worth nA can finance the n projects if and only if

$$p_H\left[R - (1 - d_n)\frac{B}{\Delta p}\right] \geqslant I - A. \tag{4.4}$$

where

$$d_n = \frac{p_L(p_H^{n-1} - p_L^{n-1})}{p_H^n - p_L^n}$$

increases with n (note that $d_1 = 0$). In the limit as n tends to infinity, d_n converges to p_L/p_H and the financing condition converges to

$$p_H R - B \geqslant I - A. \tag{4.5}$$

That is, in the limit the pledgeable income per project is equal to $p_H R - B$. Intuitively, with a large number of independent projects, shirking on a non-negligible fraction of projects is necessarily detected by the law of large numbers. And so the highest rent that the entrepreneur can grab is her private benefit B on each project.

In this model, increasing the number of projects raises the pledgeable income per project and alleviates incentive problems, but does not fully eliminate credit rationing. Recall that positive-NPV projects satisfy $p_H R \geqslant I$ and that we assumed that

$$p_H R - I < B.$$

For a given total net worth of the borrower, her net worth per project A tends to 0 as n tends to infinity and thus (4.5) is violated. In other words, a borrower with a finite net worth cannot undertake an arbitrarily large number of positive-NPV projects. Thus net worth still plays a role even with a large number of projects.

In contrast, Diamond (1984) showed that a borrower who can avail herself of a large number of projects is never credit rationed, and thus faces no

capital (or leverage) requirement. Where does this discrepancy in results come from? Here the borrower can always divert nB in private benefits. So, her rent necessarily grows proportionally with the number of projects.

Alternatively, we could have assumed away private benefits and called B the disutility of working on a project, with the disutility of shirking being normalized at 0. In this "Diamond formulation," a project has positive NPV if $p_H R \geqslant I + B$, as the disutility of effort must be counted as a cost of the project. (In contrast, in the basic formulation the borrower does not take her private benefit.) The incentive conditions remain the same as in the private benefit model, and thus the only difference between the two formulations is the definition of a positive-NPV project. Condition (4.5) then shows that in the Diamond formulation, the borrower can undertake an arbitrarily large number of positive-NPV projects provided that her cash on hand is nonnegative.

This unboundedness and the related lack of capital requirement differentiate the Diamond formulation from the one considered here. But the main message—diversification boosts borrowing capacity—is the same in both formulations.

Remark (optimality of the standard debt contract). Diamond shows that a *debt contract* with investors achieves the social optimum with a large number of projects. Suppose (somewhat informally) that in our formulation (i) there is a continuum of independent projects and (ii) $p_H R - I > B$, so we are in a situation in which the borrower can undertake an infinite number of projects without any initial net worth. Assume indeed that the borrower has no initial net worth ($A = 0$), and let the borrower issue a debt contract in which she must reimburse $D = I$ (we normalize the mass of projects to one). Investors are willing to purchase this debt claim if and only if the probability of default is equal to 0.

Let us first check that the borrower prefers behaving on all projects to shirking on all. The "law of large numbers"[6] implies that the firm's total income is $p_H R$ in the former case and $p_L R$ in the latter case. As $p_H R > I > p_L R$, the borrower's residual claims are $p_H R - I$ and 0, respectively. So, the borrower prefers working on all projects if and only if $p_H R - I > B$, which we have assumed in order to guarantee that the borrower needs no capital to undertake a large number of projects.

More generally, it is easy to check that the borrower does not benefit from working on a fraction of projects and shirking on the remaining fraction. Suppose the borrower works on a fraction κ of projects. Either $\kappa p_H R + (1-\kappa) p_L R < I$, and then there is default and the borrower would be better off shirking on all projects; or $\kappa p_H R + (1-\kappa) p_L R \geqslant I$, and then

$$\frac{\mathrm{d}}{\mathrm{d}\kappa}[\kappa p_H R + (1-\kappa)(p_L R + B)] = (\Delta p) R - B > 0,$$

and so, if $\kappa < 1$, the borrower, who receives the firm's incremental income once debt is fully reimbursed, is better off increasing κ.

The logic of the argument is clear: a debt contract makes the borrower *residual claimant* of profits whenever there is no default. So she has proper incentives to work as long as she does not choose to default (we employ "choose" on purpose, because the law of large numbers implies that there is no surprise as to whether default occurs).

4.2.3 Limits to Diversification

While the point that diversification can alleviate incentive problems and lower capital requirements is an important one, it should be realized that there are in practice a number of obstacles to diversification.

Endogenous correlation. The key to the diversification argument is that projects are independent, so that if one fails another is still likely to succeed and the latter's income is thus good collateral for the former. An important implicit assumption of the diversification argument is that the borrower cannot alter the independence through project choice; for, the borrower has an incentive to choose correlated projects ("asset substitution"). Intuitively, the correlation destroys the value of "collateral," and cross-pledging then is useless.

To illustrate this, consider the contract obtained in the case of two projects

$$\{R_2 = 2B/[(\Delta p)(p_H + p_L)],\ R_1 = R_0 = 0\}.$$

Suppose that the manager can choose two independent projects or two perfectly correlated projects,

6. Interpreted very loosely. See Diamond (1984) and Hellwig (2000) for more careful treatments, with a finite number of projects going to infinity.

but that the investors are unable to tell whether the projects are independent or correlated. By choosing correlated projects rather than independent ones, the borrower obtains

$$U_b^c = p_H R_2 > U_b^i = p_H^2 R_2,$$

where "c" stands for "correlated projects" and "i" for "independent projects," and so diversification does not occur.

This point, which is related to the discussion of "asset substitution" in Chapter 7, should not surprise the reader. The borrower's claim is an equity claim, and is therefore convex in realized income. The borrower's incentive structure makes her risk loving (even though her intrinsic preferences exhibit risk neutrality). Under correlation, the probabilities of 2, 1, and 0 successes are $(p_H, 0, 1 - p_H)$, while they are $(p_H^2, 2p_H(1 - p_H), (1 - p_H)^2)$ in the case of independent projects. Correlation therefore induces a mean-preserving spread of the distribution. And, as is well-known, risk lovers benefit from a mean-preserving spread.

Similarly, consider Diamond's debt contract, which, recall, implements the optimum with a large number of projects. Assume again that the borrower can choose between independent projects and correlated projects. Then

$$U_b^c = p_H(R - I) > U_b^i = p_H R - I,$$

so the borrower prefers correlation.

The theoretical concern expressed here underlies much of corporate risk management and of prudential reforms attempting to measure a bank's "value at risk." The covariance among activities of a firm or of a financial institution such as a bank, or of a division thereof, is often hard to measure. Financial innovation, in particular the development of derivatives, such as swaps, futures, and options, has created new opportunities for insurance against external shocks (such as interest rate or exchange rate shocks). This in principle should alleviate incentive problems by protecting managers from shocks they have no control over and thereby making them more accountable.[7] On the other hand, derivatives and other financial products can be used in the opposite direction to *increase* rather than decrease risk; and it often proves difficult for outsiders to estimate the risk pattern of a firm's or a division's portfolio. Consequently, boards of directors or chief executive officers are concerned about a division or a trader losing fortunes through nondiversified portfolios. Similarly, bank depositors (or rather their representatives, namely, the banking supervisors) are worried about failure of nondiversified banks and have been actively designing methods for measuring the riskiness of a portfolio so as to better tailor capital requirements to this riskiness.

Core business competency. Another obvious obstacle to diversification is that the borrower often has expertise only in limited sectors. Expanding within the realm of the core business may not substantially improve diversification as new activities are subject to the same industry-wide shocks as existing ones. On the other hand, diversification outside the core business activities generates inefficiencies (which can easily be modeled in our framework by introducing, say, new and independent projects with increasing stand-alone capital requirements).[8] In such situations, diversification need not boost debt capacity.

Limited attention. To the extent that diversification goes together with an increase in the number of projects, there is some concern that the borrower cannot handle that many projects. The borrower can, of course, expand and delegate the supervision of these projects to other agents, but this introduces further agency problems. Therefore, there exists a cost to diversifying through expansions.[9]

Remark (the diversification discount). A number of empirical studies, starting with Wernerfelt and Montgomery (1988), have shown that diversification

7. See Holmström (1979), Shavell (1979), as well as Section 3.2.6. Loosely stated, the "sufficient statistic theorem" states that an agent's reward should depend only on variables over which she has control.

8. A number of observers believe that the diversification of the U.S. Savings and Loans away from residential mortgages and toward commercial real estate, instalment loans, credit card loans, and corporate securities increased rather than decreased their probability of failure (this diversification was allowed by regulators in the early 1980s in response to the serious hardships then faced by the S&Ls).

9. There is a large literature on the "span of control" and the incentive cost associated with bigger hierarchies. See, for example, Calvo and Wellisz (1978, 1979), Aghion and Tirole (1997), and the references therein.

is associated with low firm value. This observation raises questions about the direction of causality (is diversification the cause of the diversification discount?) and, relatedly, as to why diversification is still so widespread despite the popularity of refocusing. Is diversification the outcome of inefficient empire building and, if so, why are boards of directors and shareholders so complacent toward managerial recommendations in this respect? Or do diversified firms simply differ from specialized ones in a number of characteristics, as several studies have indicated? For example, Villalonga (2004a,b) shows that diversified firms are present in industries with a low Tobin's q[10] and have a lower percentage of their stock owned by institutions and insiders; she argues that the diversification discount cannot be attributed to diversification itself.

We have little to say about the possibility of empire building at this stage of the book.[11] More generally, the Diamond argument is too simplistic to address the empirical evidence regarding diversification; yet it is interesting to look at its consequences. Its logic implies that it is silent about the return expected by uninformed investors: the latter receive the market rate of return regardless of the entrepreneur's diversification decision. So a diversification discount, if any, must apply to total investor shares, which in this barebones model, also include the entrepreneur's shares (or insiders' and informed investors' shares in a broader model). Consider moving from one project to two in the model above. There are several reasons why the added project may reduce profitability: the second project may have a lower return than the first (for instance, the payoff in the case of success is lower: $R_2 \leqslant R_1$; this is the core business competency argument); the avoidance of asset substitution requires costly monitoring (endogenous correlation argument); or the second project may divert managerial attention from the first (limited attention argument). In each case, the second project reduces average profitability, and yet the entrepreneur may want to undertake it if she has enough funds or the agency cost is low enough.[12] While this exercise shows how a diversification discount may arise from corporate heterogeneity rather than a poor investment pattern, it is somewhat unsatisfactory as it misses the broader discussion of the various relevant dimensions of heterogeneity that would be needed for both a better theoretical understanding of diversification and a more structured estimation of the discount and its underpinnings.

4.2.4 Sequential Projects: The Build-up of Net Worth

Section 4.7, in the supplementary section, investigates the case of a sequence of two projects, project 1 at date 1 and project 2 at date 2.[13] The key difference with the case of two "simultaneous projects" analyzed in Section 4.2.1 is that the outcome (success or failure) in the first project is realized before the investment in the second project needs to be sunk. The new feature is that the investment in the second project can be made contingent on the first project's outcome. In particular, the optimal contract may threaten the entrepreneur with nonrefinancing if the first project fails even though the projects are independent and so there is no learning about the second project's profitability from first-period performance. In the (constant-returns-to-scale) variable-investment context, the main results of that section can be summarized in the following way:

(i) The entrepreneur cannot do better through long-term contracting than entering a sequence of short-term contracts in which the investors are reimbursed only on the current project and break even in each period (no cross-pledging). The entrepreneur receives nothing and does not invest in the second project if she fails in the first period.

10. Tobin's q is equal to the market value of a firm's assets divided by the replacement value of these assets.

11. Managerial rents do grow with firm size in our model, suggesting that the borrower would push for a larger empire. The question is therefore why investors would let the borrower sacrifice investor value to increase her own managerial rent. In Chapter 10, we will discuss reasons why managers often get their own way.

12. It is, furthermore, easy to build examples in which diversified firms have a lower percentage of stocks held by insiders (due to the fact that they have to borrow more).

13. The analysis carries over to an arbitrary number of projects.

(ii) The first-period investment is larger than it would be in the absence of a follow-up project. The threat of not being able to finance the second project acts as disciplining device and alleviates date-1 moral hazard. Put differently, the fact that \$1 of entrepreneurial net worth at date 2 is worth more than \$1 to the entrepreneur due to credit rationing makes the entrepreneur more eager to behave at date 1.

(iii) Stakes are increasing: the date-2 investment in the case of date-1 success is larger than the date-1 investment.

(iv) The entrepreneur has a higher utility in the sequential-project case, as the lower agency cost boosts borrowing capacity.

(v) Project correlation need no longer reduce the entrepreneur's utility due to a learning effect: the second-period project's dimension can be made contingent on the first-period outcome, which is then informative about the date-2 prospects.

4.3 Boosting the Ability to Borrow: The Costs and Benefits of Collateralization

In the previous sections, "assets" or "net worth" referred to some form of cash that the borrower was able to put up front to defray part of the cost of investment. Some other assets cannot be used up front to participate in the financing, and yet are "quasi-cash." Suppose for instance that the entrepreneur has no cash but, as a leftover of a previous activity, will deliver some accounts receivables to a buyer, resulting for the entrepreneur in riskless profit A. So total profit will be $R + A$ in the case of success of the current project and A in the case of failure. Obviously, the entrepreneur can pledge this riskless profit A to the lenders, and everything is *as if* the entrepreneur had cash A today. Or, to emphasize the same point, suppose that the entrepreneur has no cash today, but that the investment I is used to purchase equipment or commercial real estate, that is used for the project and will after completion of the project be resold at some riskless price A. This resale value can be pledged as collateral to the lenders and is quasi-cash.

More generally, the ability to pledge productive assets may help raise external finance. This section makes a few points concerning the link between collateral and loan agreements.

4.3.1 Redeployability

We start with the straightforward point that the option to use a productive asset for other purposes outside the firm helps raise external finance. Suppose that we extend the fixed-investment framework of Section 3.2 to allow for the possibility of learning that the investment could have superior alternative uses. More precisely, let I be spent to purchase some productive asset such as land or equipment. After the investment is sunk but before the entrepreneur starts working on the project, a public signal accrues that indicates whether the project is viable:

- with probability x, the project is viable and its characteristics are as described in Section 3.2 (so, the model of Section 3.2 corresponds to $x = 1$);
- with probability $1 - x$, the parties learn that the project will not deliver any income (at least under current management), regardless of the entrepreneur's effort (for example, there might turn out to be no demand for the corresponding product or perhaps the entrepreneur will prove to be an incompetent manager of the assets).

In the second situation, labeled "distress," the asset can be sold to a third party at some exogenous price $P \leqslant I$ (this value of collateral in the case of distress is here taken as exogenous: see the discussion below). A high resale price P corresponds to a highly redeployable asset. By contrast, a specialized asset should fetch a low resale price. Commercial real estate is one of the most redeployable assets, even though resale implies a loss. At the opposite extreme lie highly specific investments such as a die (or, more generally, custom-made equipment) or the personnel's human capital investment into the project. Some equipment with well-organized second-hand markets, such as buses and airplanes, may lie in between.

The timing of this extension of the basic model is summarized in Figure 4.1.

With a positive probability of distress ($x < 1$) and with asset specialization ($P < I$), the condition for a

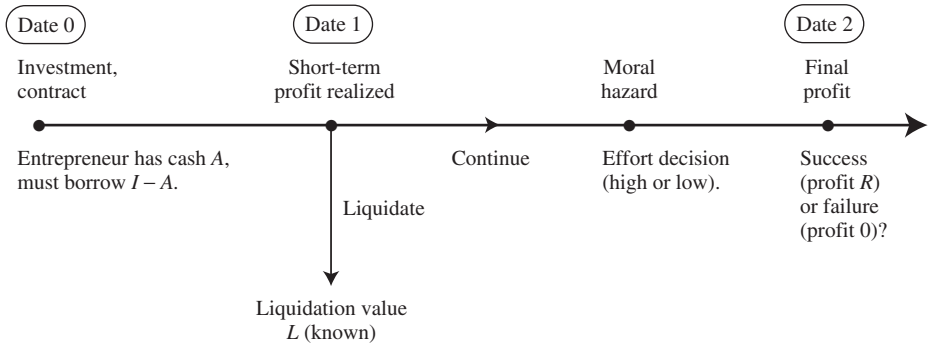


Figure 4.7

firm continues, its prospects improve with r (so r is "good news" about the future). Namely, the probability of success is $p_H(r)$ if the entrepreneur works between dates 1 and 2 and $p_L(r)$ if the entrepreneur shirks. Assume that $p'_H > 0$, $p'_L > 0$, and

$$p_H(r) - p_L(r) \equiv \Delta p$$

is independent of r (so shirking reduces the probability of success by a fixed amount independent of prospects). As usual, one will want to induce the entrepreneur to work if continuation obtains. It is convenient to use the notation

$$\rho_1(r) \equiv p_H(r)R \quad \text{and} \quad \rho_0(r) \equiv p_H(r)\left[R - \frac{B}{\Delta p}\right].$$

Investors are competitive and demand an expected rate of return equal to 0. Assume

$$\rho_1(r) > L \quad \text{for all } r \tag{1}$$

and

$$E[r] + L > I - A > E[r + \rho_0(r)]. \tag{2}$$

(i) Argue informally that, in the optimal contract for the borrower, the short-term profit and the liquidation value (if the firm is liquidated) ought to be given to investors.

Argue that, in the case of continuation, $R_b = B/\Delta p$. (If you are unable to show why, take this fact for granted in the rest of the question.)

Interpret conditions (1) and (2).

(ii) Write the borrower's optimization program.

Assume (without loss of generality) that the firm continues if and only if $r \geqslant r^*$ for some $r^* \in (0, \overline{r})$. Exhibit the equation defining r^*.

(iii) Argue that this optimal contract can be implemented using, inter alia, a short-term debt contract at level $d = r^*$. Interpret "liquidation" as a "bankruptcy."

How does short-term debt vary with the borrower's initial equity? Explain.

(iv) Suppose that, when the decision to liquidate is taken, the firm must go to a bankruptcy court. The judge mechanically splits the bankruptcy proceeds L equally between investors and the borrower.

Define $\hat{r}$ by

$$\rho_0(\hat{r}) \equiv \tfrac{1}{2}L.$$

Assume first that

$$r^* > \hat{r}$$

(where r^* is the value found in question (ii)).

Show that the borrower-friendly court actually prevents the borrower from having access to financing. (Note: a diagram may help.)

(v) Continuing on question (iv), show that when

$$r^* < \hat{r},$$

the borrower-friendly court either prevents financing or increases the probability of bankruptcy, and in all cases hurts the borrower and not the lenders.

Exercise 4.10 (benefits from diversification with variable-investment projects). An entrepreneur has two variable-investment projects $i \in \{1, 2\}$. Each is described as in Section 3.4. (For investment level I^i, project i yields RI^i in the case of success and 0 in the case of failure. The probability of success is p_H if the entrepreneur behaves (and thereby gets no private benefit) and $p_L = p_H - \Delta p$ if she misbehaves (and then obtains private benefit BI^i). Universal risk neutrality prevails and the entrepreneur is protected

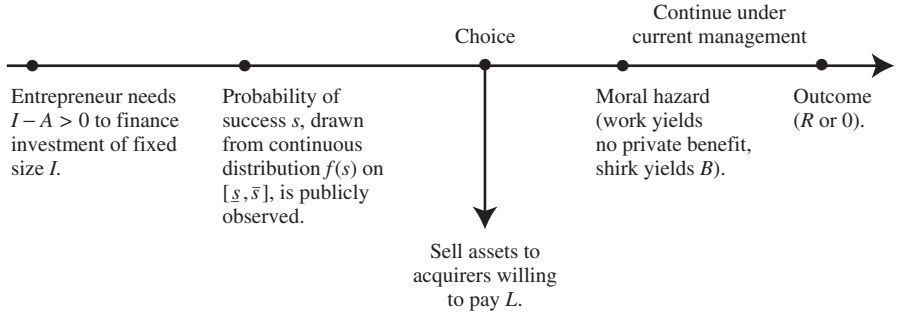


Figure 4.8

by limited liability.) The two projects are *independent* (not correlated). The entrepreneur starts with total wealth A. Assume

$$\rho_1 \equiv p_{\mathrm{H}} R > 1 > \rho_0 \equiv p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right)$$

and

$$\rho_0' \equiv p_{\mathrm{H}}\left(R - \frac{p_{\mathrm{H}}}{p_{\mathrm{H}} + p_{\mathrm{L}}}\frac{B}{\Delta p}\right) < 1.$$

(i) First, consider project finance (each project is financed on a stand-alone basis). Compute the borrower's utility. Is there any benefit from having access to two projects rather than one?

(ii) Compute the borrower's utility under cross-pledging.

Exercise 4.11 (optimal sale policy). Consider the timing in Figure 4.8.

The probability of success s is not known initially and is learned publicly after the investment is sunk. If the assets are not sold, the probability of success is s if the entrepreneur works and $s - \Delta p$ if she shirks (in which case she gets private benefit B). Assume that the (state-contingent) decision to sell the firm to an acquirer can be contracted upon *ex ante*. It is optimal to keep the entrepreneur (not sell) if and only if $s \geqslant s^*$ for some threshold s^*. (Assume in the following that s has a wide enough support and that there are no corner solutions. Further assume that, conditional on not liquidating, it is optimal to induce the entrepreneur to exert effort. If you want to show off, you may derive a sufficient condition for this to be the case.) As is usual, everyone is risk neutral, the entrepreneur is protected by limited liability, and the market rate of interest is 0.

(i) Suppose that the entrepreneur's reward in the case of success (and, of course, continuation) is $R_{\mathrm{b}} = B/\Delta p$. Assuming that the financing constraint is binding, write the NPV and the investors' break-even constraint and show that

$$s^* = \frac{(1+\mu)L}{R + \mu(R - B/\Delta p)}$$

for some $\mu > 0$. Explain the economic tradeoff.

(ii) Endogenize $R_{\mathrm{b}}(s)$ assuming that effort is to be encouraged and show that indeed $R_{\mathrm{b}}(s) = B/\Delta p$ for all s. What is the intuition for this "minimum incentive result"?

(iii) Suppose now that s can take only two values, s_1 and s_2, with $s_2 > s_1$ and

$$s_2\left(R - \frac{B}{\Delta p}\right) > \max(L, I - A).$$

Introduce a first-stage moral hazard (just after the investment is sunk). The entrepreneur chooses between taking a private benefit B_0, in which case $s = s_1$ for certain, and taking no private benefit, in which case $s = s_2$ for certain. Assume that financing is infeasible if the contract induces the entrepreneur to misbehave at either stage. What is the optimal contract? Is financing feasible? Discuss the issue of contract renegotiation.

Exercise 4.12 (conflict of interest and division of labor). Consider the timing in Figure 4.9.

The entrepreneur (who is protected by limited liability) is assigned two simultaneous tasks (the moral-hazard problem is bidimensional):

- The entrepreneur chooses between probabilities of success p_{H} (and then receives no private benefit) and p_{L} (in which case she receives private benefit B).
- The entrepreneur is in charge of overseeing that the asset remains attractive to external buyers